Tutorial Lab 12.7 (Formatively Assessed):

Corner Analysis

Preliminary Matter: Setting up the Cadence Environment

Please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout before entering these commands into your terminal.

For this course, you will use the Cadence Virtuoso Design Environment (referred to simply as Cadence). Cadence is a Linux based software suite for designing full custom integrated circuits. In order to use Cadence you will first need to set-up the Cadence environment.

8. Open a terminal window and ensure that you are in your home directory using the following command:

```
cd ~
```

Note: 'cd' stands for change directory. The solitary tilde symbol (~) is a shortcut/alias for your home directory (/home/username).

9. Under your home directory, use the following commands to load the Cadence environment:

```
module use /eda/cadence/modules

module load course/EENGM6030
```

Note: The two 'module' commands load the required shell/Cadence environmental variables for this course (paths, names, aliases…) In particular, this includes a reference to the process design kit (PDK) used in this course, the GPDK180. You will need to do this every time you open a terminal, since the sourced environment only applies to the current terminal (until it's closed).

a. ***Note: This step only needs to be completed ONCE - If you already ran this script in lab 12.1, you don't need to run it again here.*** The first time you use Cadence, you will need to run the following command (script):

```
EENGM6030_setup
```

This will create a folder called AICD (short for Analogue Integrated Circuit Design) under your home directory. The AICD folder contains the files and directories needed to run Cadence. Running this setup script more than once will result in a warning message that the AICD folder already exists. It will then exit without making changes.

10. The AICD folder will be used for all your Cadence related activities for this course. **You should always make sure you are in your AICD folder before you launch Cadence.** To run cadence, use the following commands:

```
cd ~/AICD

virtuoso &
```

.

1. Learning Objectives of this Lab:

From the previous lab, you should be able to:

- Create, and simulate in terms of a DC-sweep, a CMOS differential voltage amplifier with resistive and current source loads.
- Understand the need for feedback in order to set the DC operating point (common-mode output voltage) of a fully differential amplifier.
- Understand the logical parts and basic arrangement of the common-mode feedback loop. Simulate a simple common-mode feedback loop using ideal components.

In this lab, you will:

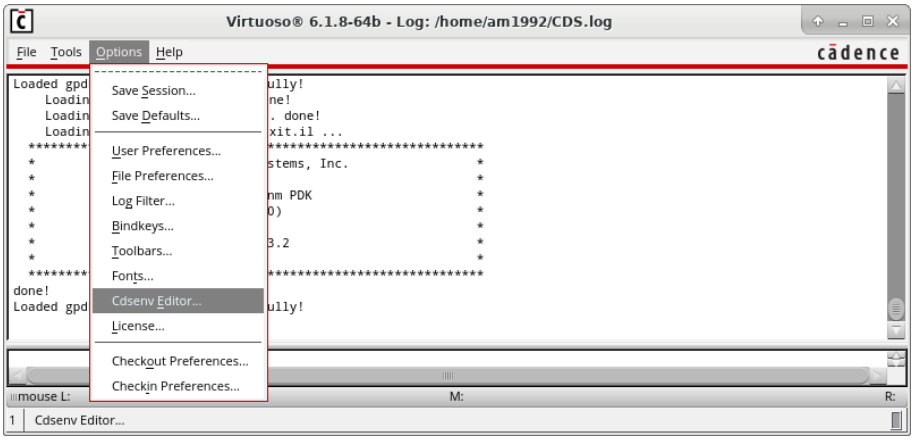
- Undertake a circuit operating point analysis over a temperature range.
- Use a parametric analysis to investigate the operation of a circuit at nominal and extremes of voltage supply.
- Simulate a CMOS FET circuit at its process corners.
- Examine the use of feedback to remove the problems associated with PVT variation.

2. Cadence Documentation, Help, and Licensing

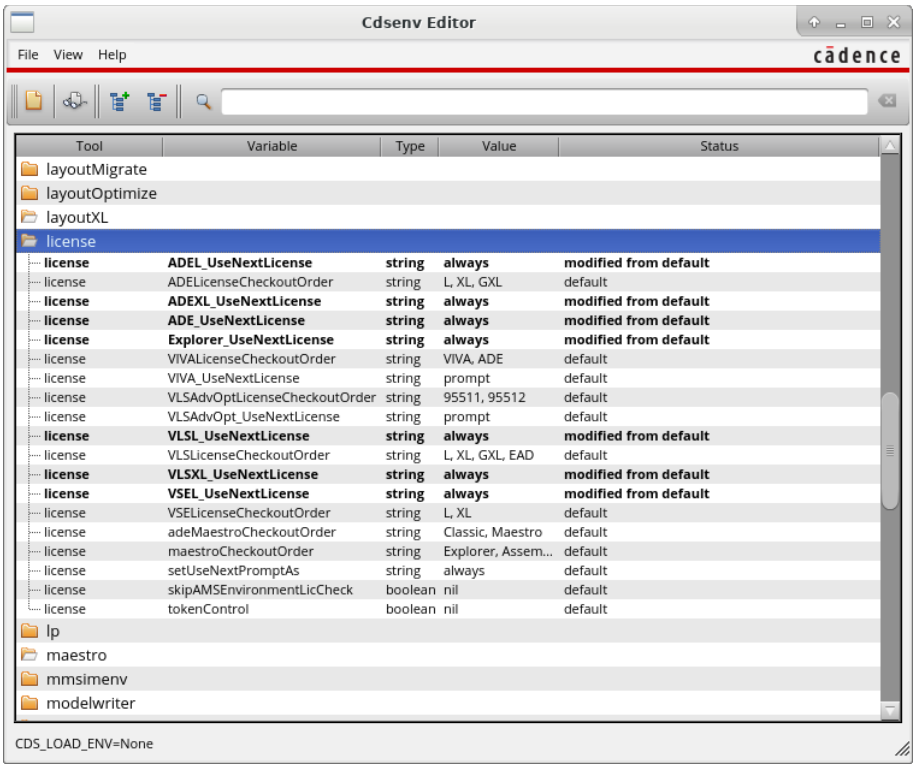
Local and online help is available from the help menus (**Menu: Help → User Guide** or by pressing **F1**), or by typing 'cdnshelp &' at the command prompt (assuming you have setup your terminal environment).

PLEASE NOTE: WHEN LAUNCHING A TOOL IN THE CADENCE DESIGN ENVIRONMENT YOU MAY SOMETIMES GET A WARNING ABOUT A LICENSE BEING UNAVAILABLE FOR THE TOOL, AND REQUESTING PERMISSION TO OBTAIN A HIGHER-TIERED LICENSE. YOU SHOULD SELECT 'ALWAYS' TO THESE MESSAGES.

If you accidently deny a license request, or you cannot access a license, you will need to modify the Cadence environment licence/permissions from the CIW. From the CIW menu go to **Menu: Options → Cdsenv Editor**. Once in the environment editor, go to the license folder, and make sure all licenses (in bold) are set to '**always'** – if any entries are listed as '**never**' you will need to manually change/retype them to '**always**'.



< Figure X.1. The Cadence Environment Menu Option from the CIW Window. >



< Figure X.2. The Cadence Environment Editor. Note, all licenses should be set to 'always'.>

3. Getting Started with Cadence:

For a refresher/introduction on how to start a Cadence session please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout.

3.1 Dealing with Lock files

PLEASE NOTE: IF A SCHEMATIC ISN'T PROPERLY CLOSED DOWN (FOR EXAMPLE, DUE TO A DROPPED REMOTE CONNECTION), THE SCHEMATIC MIGHT BECOME EDIT LOCKED. IF THIS HAPPENS YOU WILL NOT BE ABLE TO OPEN THE FILE FOR EDITING. YOU WILL ONLY BE ABLE TO OPEN THE FILE FOR READING UNTIL YOU REMOVE ANY/ALL LOCK FILES.

There are two ways you can unlock files:

(6) The first is to manually delete any edit lock files. To do this, go to your '~/AICD/Designs' folder and find the folder representing the schematic e.g., for lab1 this would be '~/AICD/Designs/lab121_vdivider/schematic' if this is what you called your schematic. Delete all the .cdslck lock files in this folder and you should be able to open your schematic for editing.

(7) You can also automatically release edit locks with the command:

```
clsAdminTool -are .
```

To use this command, you have to be in your AICD folder (the full-stop is shorthand for the current directory path). If you are not in your AICD folder, you have to specify the path (relative or absolute) to your AICD directory.

Since this is a Cadence command, you have to ensure that you have set up the terminal environment for Cadence (in particular, the modules must be loaded as in step 2, 'Preliminary Matter'). It is best to use the terminal window that you used to start Cadence, since the environment was already set up. Note that you will not be able enter any commands into your terminal if you did not start cadence with the '&' character. By example, to release any lock files after opening a new terminal, the full command set would be,

```
cd~
module use /eda/cadence/modules
module load course/EENGM6030_2021
clsAdminTool -are AICD
```

For more information on this command, use

```
clsAdminTool -help
```

4. Circuit Simulation Exercises: Part 1

With guidance, you will:

- Undertake a circuit operating point analysis over a temperature range.

The simulation device models (PDKs) we use in analogue CAD are complex. Underlying these models are dozens of equations using several dozen model parameters – including physical variables, environmental variables, process variables, and coefficients of operation.

When designing analogue integrated circuits, we must be aware that the model parameters used in our simulations are often initially set to their nominal/typical 'statistical mean' values. Furthermore, the device/circuit temperature is also initially set to a nominal value, typically, 18–27°C). When realised, however, the circuits we are designing will be subject to significant variation in their manufacturing and processing (e.g., doping concentration profiles, dimensional tolerances, bulk and surface-level defects, etc.), and in their operating environment (chiefly, temperature and power supply). It is realistic to expect that two circuits fabricated on the same wafer may have significantly different performance.

Simulating for only the nominal conditions of process, voltage, and temperature is not sufficient to prove functionality of your design. We must ensure that our designs will function (1) correctly and (2) within specification, over all likely variations of **p**rocess, **v**oltage, and **t**emperature (PVT). Fortunately, simulation device models include information on the statistical variation of their process parameters. We use such information during the design process using two types of analyses – **Corner Analysis** (a form of worst-case/extreme-case analysis) and **Monte-Carlo Analysis** (a form of random analysis).

Internally, device model libraries are divided into sections. In the case of the corner analysis, the worst-case variations of process parameters are grouped according to their effect on **PMOS and NMOS device speed** – specifically, carrier mobility and transconductance. These groupings are placed in separate sections within the device model library:

- **NN**. The typical mean values of the model parameters that give rise to the average speed (nominal or typical) behaviour for PMOS/NMOS devices are grouped together in a section designated **NN** (Nominal NMOS / Nominal PMOS).

- **SS**. The worst-case combination of model parameters that give rise to the slowest NMOS and slowest PMOS devices (poor mobility/transconductance) are grouped together in a section of the device model designated **SS** (slow NMOS / slow PMOS).

- **FF**. The worst-case variation of process parameters that give rise to the fastest NMOS and fastest PMOS devices (high mobility/transconductance) are grouped together in a section of the device model designated **FF** (fast NMOS / fast PMOS).

- **SF** and **FS**. There are two other worst-case parameter combinations: slow-NMOS/fast-PMOS devices (**S/F**), and fast-NMOS/slow-PMOS devices (**F/S**).

Each of the four worst-case conditions represents a 'process corner' in our simulations:

Section	NMOS	PMOS
NN	Nominal	Nominal
SS	Slow	Slow
SF	Slow	Fast
FS	Fast	Slow
FF	Fast	Fast

< Table 1. Nominal Process Parameters plus Four 'Worst-Case' Process Corners used in the GPDK180 PDK >

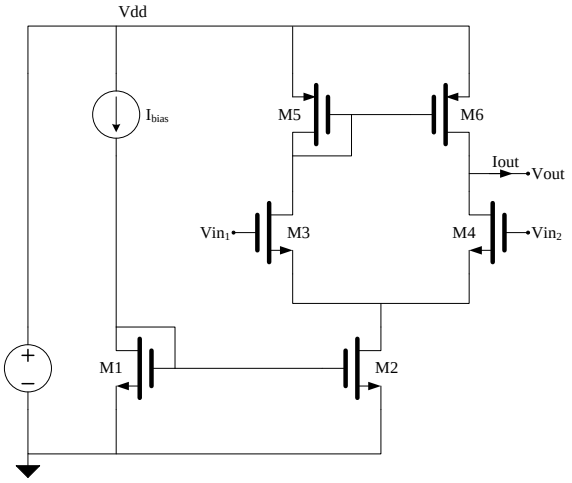
Note: Some PDK suppliers may use a different designation. For example, the AMS 0.35um PDK you used previously uses **TM** (typical-mean), **WS** (worst-case speed) and **WP** (worst-case power). In this case, **TM**=**NN**, **WS**=**SS** and **WP**=**FF** (fast devices come at the cost of using more power which is why WP=FF). Additionally, **WO**=**SF** (worst-case 'one' switching means slow PMOS) and **WZ=FS** (worst-case 'zero' switching means slow NMOS). uses this designation. The principle of table 1, however, is the same.

For a full PVT corner analysis, we must also vary the power supply and temperature in addition to the process corners, so that we can observe the ***simultaneous*** effect of process (NN, SS, FF, SF, FS), voltage (VDD +/- 10%), and temperature (-40°C to +125°C) on our circuits.

For the curious, the GPDK180 model library is based on the BSIM3v3 SPICE model (*BSIM* = **B**erkeley **S**hort-channel **I**nsulated-gate field-effect-transistor **M**odel). For a full list of the model equations and parameters you are referred to APPENDIX A of the BSIM3v3 manual https://link.springer.com/content/pdf/bbm%3A978-0-306-47050-9%2F1.pdf.

4.1 The Differential Transconductance Amplifier (TCA)

(4.1.1) Create the schematic for the NFET-inut differential transconductance amplifier in Figure 1. A summary of the circuit's operation is given in section 4.4 of the background material.

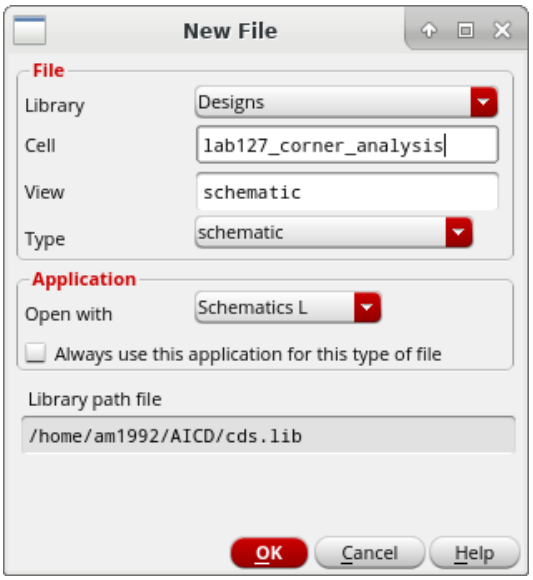


< Figure 201. Current Mirror Loaded, NFET Input, Differential Transconductance Amplifier (TCA). Reproduced from Figure 26, section 4.4 of the background material. >

(4.1.2) From the Library Manager window, select the 'Designs' library and choose **Menu: Files à New à Cell View** to invoke the New File form (Figure 2).

In the 'Cell' text field, choose a meaningful cell name (e.g., lab127_corner_analysis). In the 'View' text field, type 'schematic' as the view name and from the 'Type' drop-down menu choose 'schematic' as the view type. In the 'Application' section of the dialog box, choose 'Schematics L' from the 'Open with' drop down menu. OK the form.

(4.1.3) You should be presented with a blank **Virtuoso Schematic Editor L** schematic entry window for the new cellview. This cellview will not be saved in the Library Manager until you check and save it in the Schematic Editor. From the Schematic Editor window, choose **Menu: File à Check and Save**. (Or you can also **Menu: Check à Current_Cellview** followed by **Menu: File à Save** to perform the same command.)



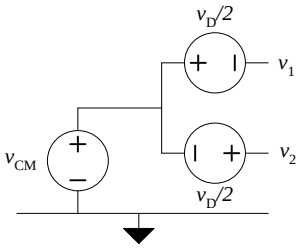
< Figure 202. The New File Form >

(4.1.4) In your schematic editor window, from the GPDK180 library, instantiate and place four **nmoshv** NMOS FETs (two for the input pair, and two for the current mirror tail bias) and two **pmoshv** PMOS FETs (for the active current-mirror load).

*Recall, there are several ways of rotating/flipping a part. When adding an instance to your schematic you can rotate and/or flip the item before you place it on the canvas using the **Rotate**, **Sideways**, and **Upside-Down** buttons on the Add Instance form. Alternatively, you can rotate/flip the components after placing them by selecting the component and using the **rotate** or **flip** commands from the menu/toolbar. You can also flip or rotate a component by using **F3** with the **move/stretch** commands. In general, pressing F3 while a command is in progress gives you a more detailed set of options for that command. When moving an item using the **Move** or **Stretch** commands, pressing F3 will bring up a dialog box with more detailed options including **Rotate**, **Sideways**, and **Upside-Down** buttons.*

(4.1.5) Instantiate and place an ideal **vdc** source for the supply voltage and an ideal **idc** current source for the tail current reference. All ideal components are from the analogLib library.

(4.1.6) To drive the circuit, use a single **vdc** for the common-mode input voltage (VCM) and add two split **vdc** inputs for to the differential-mode input voltage (vd). This will enforce the common-mode voltage (CMV) principle rather than having separate VCMs driven by the same parameter for the two inputs (**note the opposing orientation of the two v_D/2 supplies – if you orient them in the same direction make one of the supplies negative, -vd/2**). See Figure 3.



< Figure 203. Common-mode voltage driver circuit.

v_1 = VCM - vd/2. v_2 = VCM + vd/2. v_D = (v_1 - v_2). VCM = (v_1 + v_2)/2.>

(4.1.7) Parameterise your design. Set the tail current reference to 100 uA (**DC Current** = 100u) For the DC supply voltage, set the voltage attribute to the string **DC Voltage** = VDD. For the input set the common-mode voltage component to the string **DC Voltage** = VCM, and the two differential-mode components to the string **DC Voltage** = vd/2 and - vd/2. Dimension all PMOS and NMOS devices 10um/1um (**width** = *10u*, **length** = *1u*). Leave all other attributes unchanged.

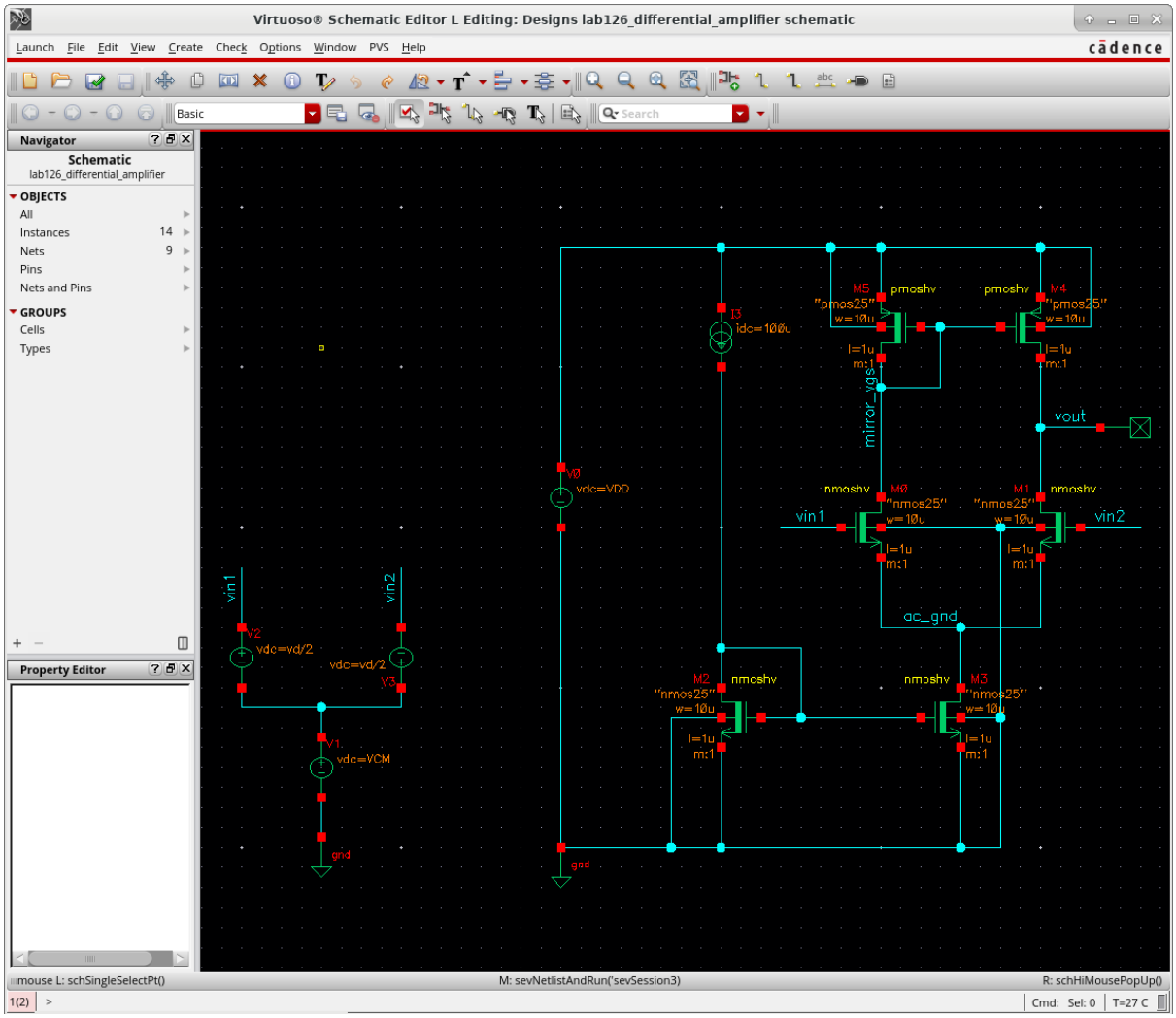
(4.1.8) Wire the components as in Figure 4 using the **Wire (narrow)** command.

(4.1.9) Using the **Create Wire Name** command: name the two input nets vin1 and vin2 (the gates of the differential input pair), name the output net vout, the drain of the mirror load mirror_vgs, and name the AC ground point ac_gnd. Use the **noConn** connector from the **basic** library to avoid floating-net warnings. Keep the schematic neat.

(4.1.10) To connect the inputs to the amplifier to the driver circuit, we will create an implicit connection using wire names. Name the outputs of the common-mode driver circuit vin1 and vin2. This will create a connection between the vin1 of the driver circuit and vin1 of the amplifier, and a connection between vin2 of the driver and vin2 of the amplifier. See Figure 4.

REMEMBER: FOR AN NMOS DEVICE, YOU MUST ALWAYS CONNECT THE BODY/BULK TERMINAL TO THE MOST NEGATIVE NET (WIRE) OF THE CIRCUIT. IN FIGURE 4, THIS IS THE GROUND TERMINAL. FOR A PMOS DEVICE, THE BODY/BULK TERMINAL MUST ALWAYS BE CONNECTED TO THE MOST POSITIVE NET (WIRE) OF THE CIRCUIT, USUALLY THE SUPPLY VDD. THIS IS NECESSARY TO ENSURE THAT THE xMOS CHANNELS WILL BE PROPERLY ESTABLISHED.

(4.1.11) Check and save your schematic using the **Check and Save** command. Remember to use the output area of the CIW to diagnose any problems with your schematic and/or simulations if there are any warning or error messages when checking, saving, or simulating your work.



< Figure 204. The Differential Transconductance Amplifier Schematic as in Figure 1. >

4.2 Simulating the Amplifier over a Temperature Range

(4.2.1) We will first simulate the DC operating point of the amplifier. Ordinarily, we would do this using the 'Save DC Operating Point' option on the DC analyses form, and then annotate the schematic. However, here we want to observe how the DC operating point varies with temperature so we will do this using a DC Sweep Analysis – with temperature as the sweep variable.

(4.2.2) From the Schematic Editor window, launch the Analogue Design Environment (ADE) – **Menu: Launch à ADE L**. Copy the design variables from the schematic into the ADE. For this circuit, these are the DC voltages from the three **vdc** components – VDD, VCM, and vd. From the ADE window, choose **Menu: Variables à Copy From Cellview**. You should now see the VDD, VCM, and vd variables. in the 'Design Variables' list of the ADE. Initialise VDD *= 2.5V*, VCM *= 1.25V,* and vd = 0V. These values are used only to set the DC operating point of the amplifier circuit.

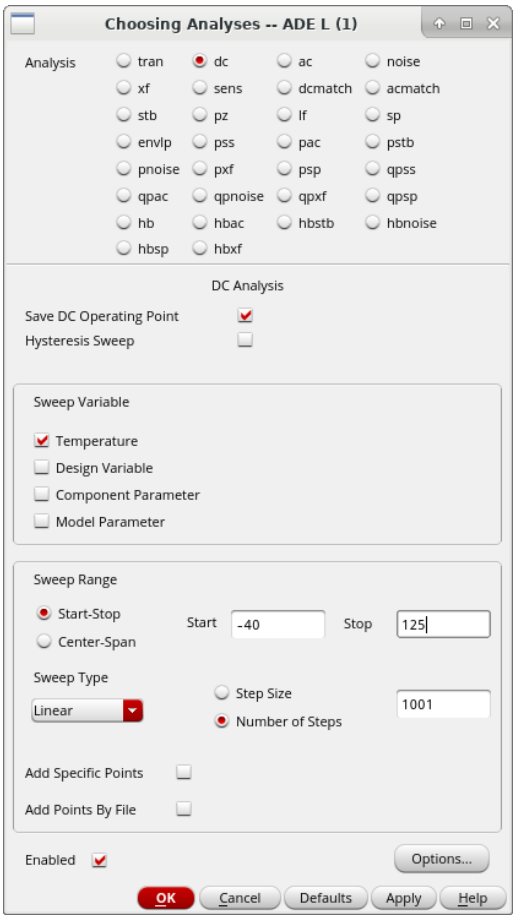
(4.2.3) We want to plot the various nets/nodes around the differential amplifier as we sweep the temperature from -40°C to +125°C. This is the full temperature range for industrial applications. See table 2.

Electronic Application	**Temperature Range (°C)**
Commercial	0 to +70
Hi-Temperature Commercial	-10 to +100
Extended Commercial	-20 to +85
Industrial	-40 to +85
Automotive	-40 to +105
Full-Range Industrial	-40 to +125
Power Supply	-40 to +130
Military	-55 to +125

< Table 2. Temperature Ranges for Different Applications. >

(4.2.4) From the ADE window, go to **Menu: Analyses à Choose…** or use the '**Choose Analyses**' button on the right-side of the ADE window (top button). The 'Choosing Analyses – ADE L' form will open. From the form, select the '**dc**' analysis option button.

In the choosing analyses form. Check the ‘Save DC Operating Point’ box. Check the ’**Temperature**’ box as the sweep variable. Choose the ‘Start-Stop’ option button with Start = -40 and Stop = 125 for the sweep range. Select ‘Linear’ from the drop-down menu as the sweep type. Choose the ‘Number of Steps’ option button with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the ‘Enabled’ box near the bottom of the form. OK the form. See Figures 5 and 6. The **dc** analysis should now appear in the ‘Analyses’ list of the ADE window.



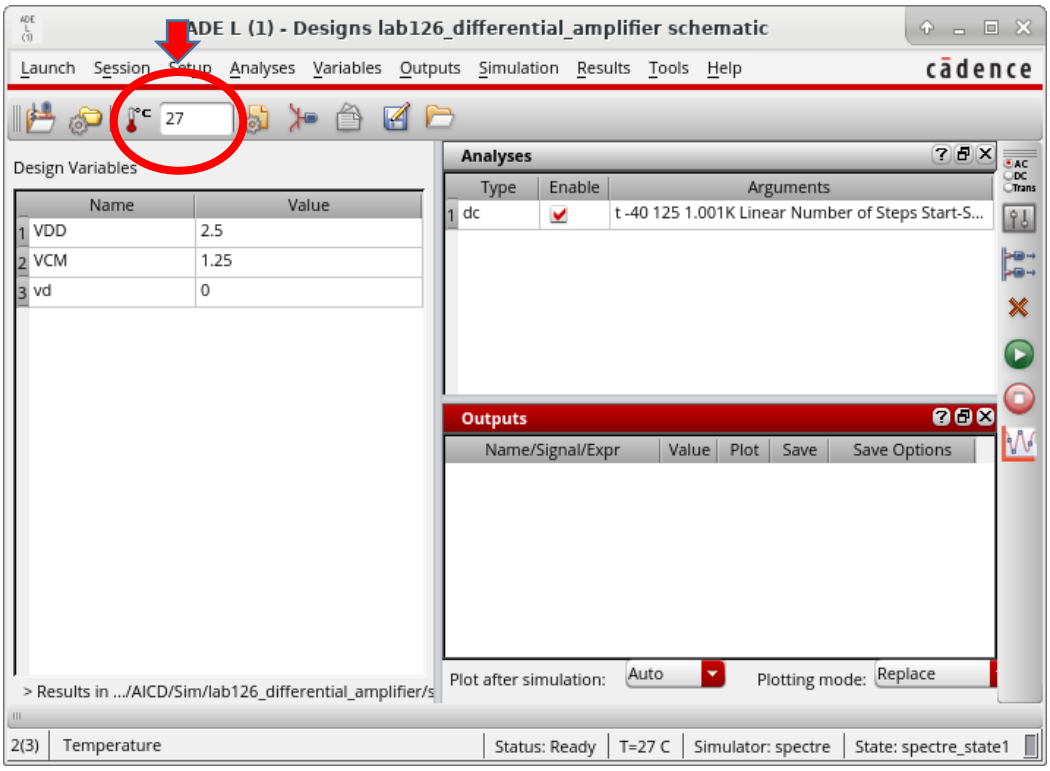
< Figure 205. The Choosing Analyses Form. >

(4.2.5) At this stage, don’t select any outputs to be plotted as we will use the Direct-Plot form instead – post simulation. In the Analog Design Environment window, run the ‘**Netlist and Run**’ command by clicking on the green play button. Only the text output window will open (since we aren’t plotting any waveforms yet).

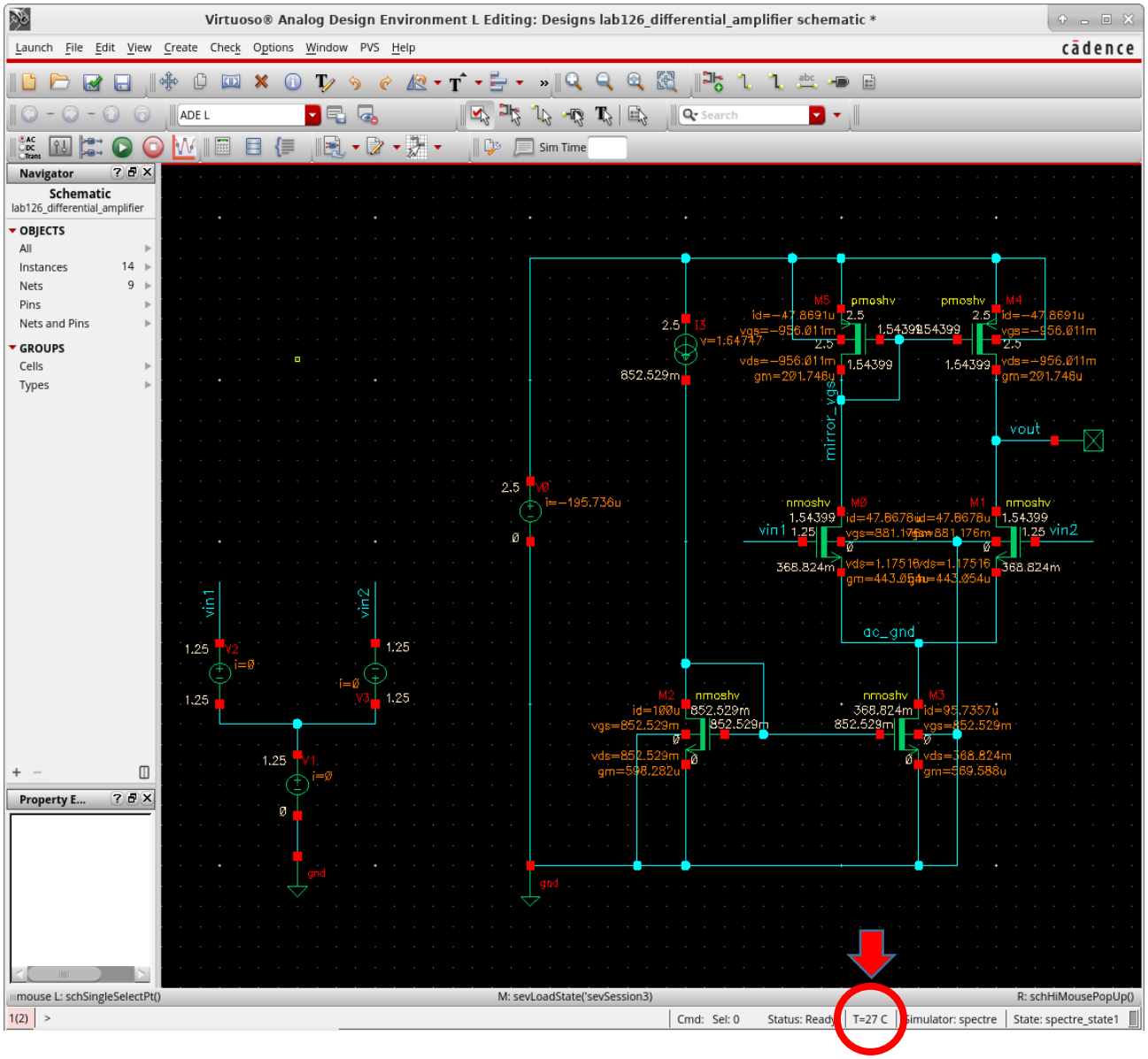
(4.2.6) Annotate the schematic, **Menu: Results à Annotate à DC node voltages…** and **Menu: Results à Annotate à DC operating points…** See Figure 7.

(4.2.7) **Discussion**. In all of the previous labs, we have simulated our circuits at a temperature of 27°C – this is the nominal temperature. This value is set in the top right of the ADE in the icon toolbar. This sets the temperature of all devices in the schematic. This will not affect the ideal components as these are not dependent on temperature. For an individual simulation you can, change the temperature on the toolbar. You can also change the temperature units (Celsius, Fahrenheit, or Kelvin) by clicking on the thermometer icon. You can also see the temperature on the status bar at the bottom of the schematic editor window.

Note that when you annotate a schematic, the annotations are for the nominal temperature (as well design variables) set in the ADE. In this case, nominal DC output operating point vout is 1.54399V (at VDD=2.5V, Temp=27°C, and Process=NN).

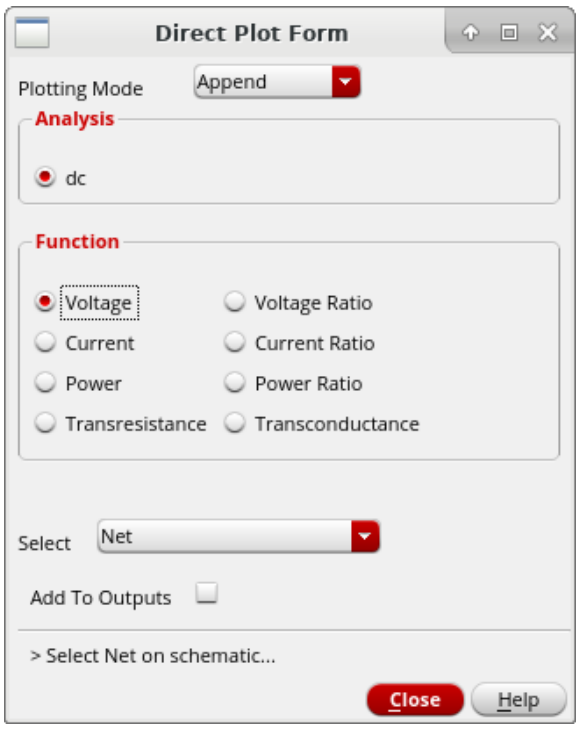


< Figure 206. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 207. Annotated. Differential Transconductance Amplifier Schematic >

(4.2.8) From the ADE window, select **Menu: Results → Direct Plot → Main Form** to invoke the Direct Plot Form (this may also invoke an empty waveform window).



< Figure 208. The Direct Plot Form for DC Analysis. >

Note: Recall, that the Direct Plot command of the ADE is a more general approach to plotting outputs after a simulation has run. Using this option, you can quickly probe the circuit – selecting and plotting the waveforms at different nets (voltages) and nodes (currents) one by one – without having to set them up in the ADE beforehand. You can also plot common expressions quickly without having to use the calculator. By default, all voltages (nets) are saved and are always available for plotting using the Direct Plot form.

Note: By default, not all currents (nodes) are saved and therefore may not be available for plotting using the Direct Plot form (MOSFET nodes are not available unless saved). To make all currents available for plotting, we can change the default behaviour of the simulator to save all device currents ***Menu: Outputs → Save All*** *and check the 'all' option in the 'Select device currents (currents)' section, or use select* ***Menu: Outputs → To Be Saved → Select On Schematic*** *and choose the nodes you want to save (you will have to re-run the simulation). These will then be available for plotting from the Direct plot form.*

Note: In the 'Analysis' section of the Direct Plot Form there will be an option button for every analysis type setup and run in the ADE. In this case, since we have only setup and run a DC analysis, there is only a ***dc*** *option present. Associated with each analysis type, will be a number of plotting options; for the ac analysis we have the function options (e.g., voltage, current, power…) and the modifier options (there are no modifier options for dc analysis - the modifier options available can be very different for the different analysis types (ac, dc, trans, stb…).*

Note: At the very top of the form is the plotting mode drop-down menu, with the usual options – Replace, Append, New Sub-Window, and New Window. Leave it on the 'Append' option.

(4.2.9) From the Direct Plot Form:

1) Select the **voltage** option in the function section of the direct plot form. Then, in the schematic window click on the output net *vout.* You should now have a plot of the dc voltage at this point over the temperature range -40°C to +125°C. See Figure 9.

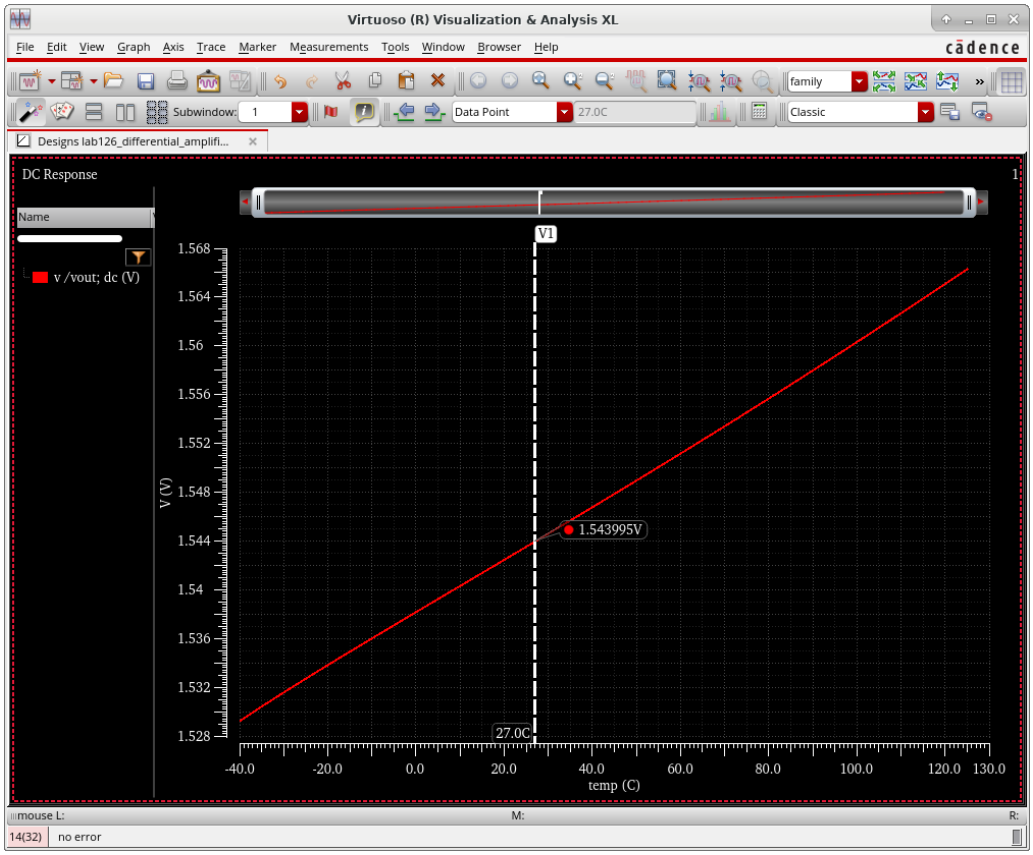
2) Select the **current** option in the function section of the direct plot form. Then, in the schematic window click on the current node for the drain of the tail device. You should now have a plot of the dc node currents over the temperature range -40°C to +125°C. See Figure 10. If the tail current is not available to plot you will need to follow the instructions above for it to be saved and re-run the simulation.

(4.2.10) From the schematic editor window, **press the 'ESC' key to exit the plotting mode of the Direct Plot Form**. Or simply close down the Direct Plot Form. Split the traces for clarity. Right-click on the empty area of the graph window and select Split Current Strip à trace from the context menu. A vertical marker has been added at the nominal temperature 27°C

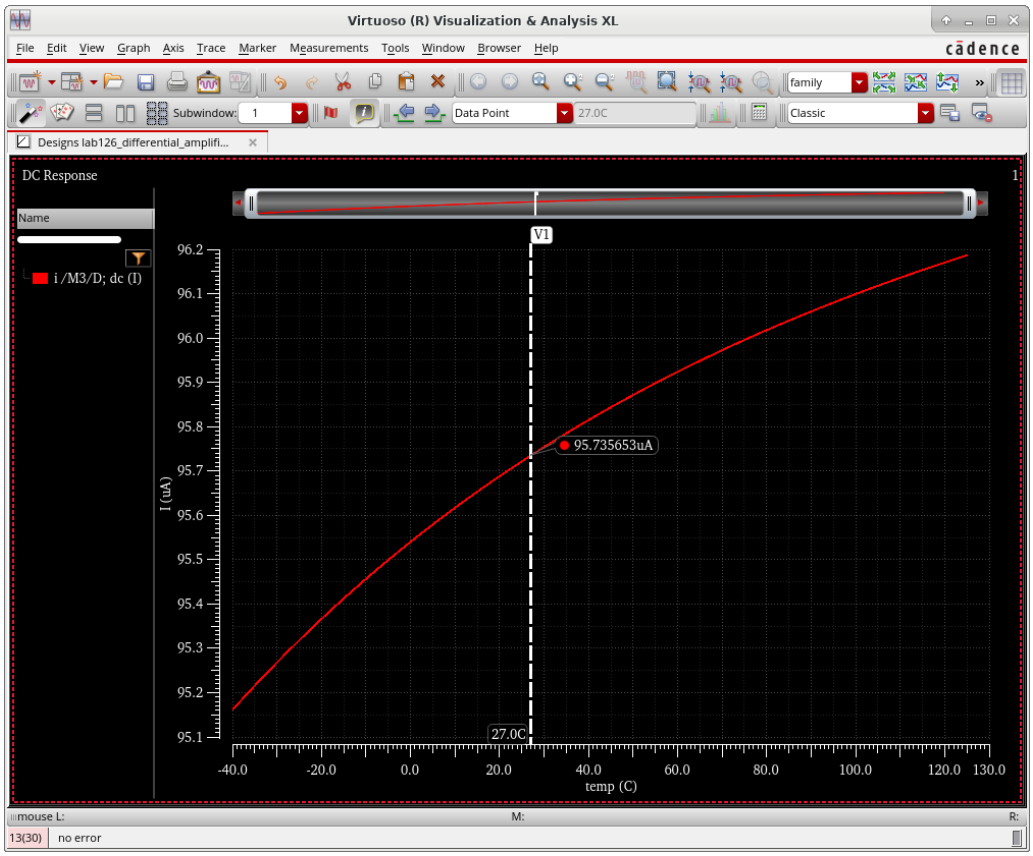
(4.2.11) **Discussion**. Referring to Figures 9 and 10. A marker has been placed on the temperature sweep at 27°C. This identifies the nominal value (1.54399V) we saw previously on our schematic annotation (see Figure 7).

For the NFET-input differential transconductance amplifier the change in operating point v_{out} over the 165°C sweep (-40 to +125°C) is around 40mV. This would not be considered significant – a change of 2 to 3% over the full range or Nominal +/- 1.5%. Similarly for the tail current, the nominal value (at VDD=2.5V, Temp-27°C, and Process=NN) is 95.736uA. This varies by +/-0.6uA over the full temperature range (a change of Nominal +/-0.6%).

This Differential Transconductance Amplifier is therefore relatively robust to temperature variation – though bear in mind this is most likely due to the large 10um/1um device sizes.



< Figure 209. DC operating point v_{out} of the TCA for temperature -40°C to +125°C >

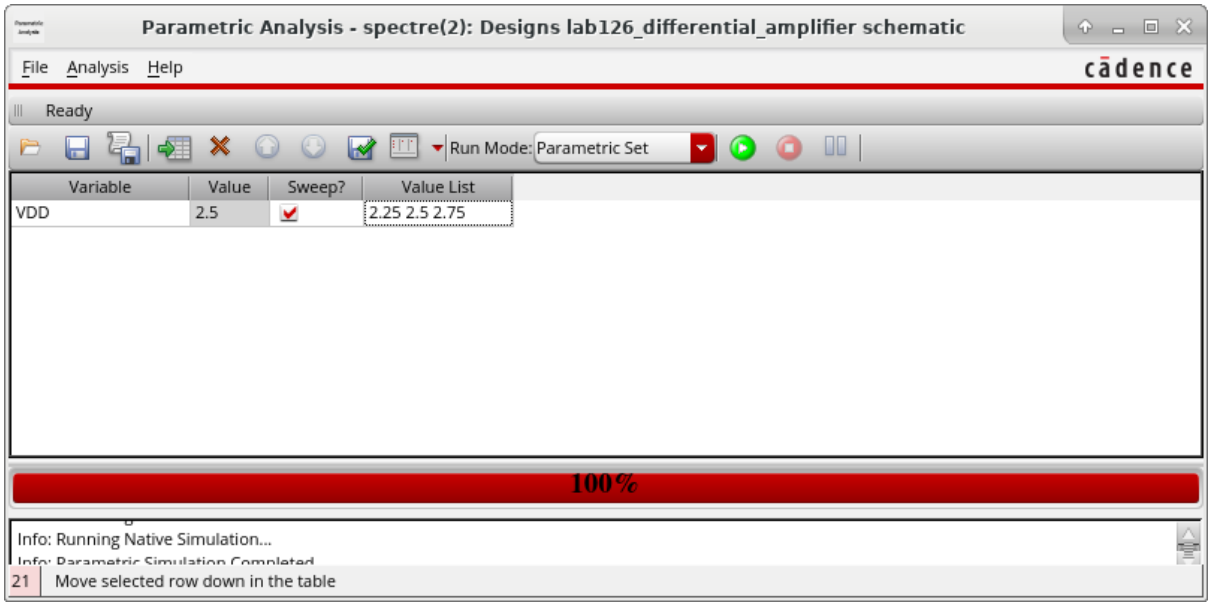


< Figure 210. DC operating point I_{TAIL} of the TCA for temperature -40°C to +125°C >

4.3 Simulating the Amplifier over a Temperature and Supply Voltage Range

(4.3.1) As well as observing the effects of varying temperature, we want to observe how our design will perform when there is a variation on the power supply voltage. In our simulations we will assume a standard VDD variation of +/-10%. We will setup a parametric analysis so we can observe the ***simultaneous*** effect of temperature and VDD variation on our circuit.

(4.3.2) To sweep VDD +/-10% we will use the parametric analysis tool. From the ADE window, choose **Menu: Tools → Parametric Analysis…** to invoke the Parametric Analysis tool. Note, that a parametric sweep can be specified as a range to sweep from the tool bar (Run Mode → Sweeps and Ranges) or as an explicit list of values over which to iterate (Run Mode → Parametric Set).



< Figure 211. Parametric Analysis Tool. For VDD = 2.5V, set up the parametric set VDD (+/-10%) = 2.25V, 2.5V, and 2.75V. This will run the swept DC-analysis (temp -40°C to 125°C) 3 times to create a family of curves. >

(4.3.3) In the Parametric Analysis window, select 'Parametric Set' from the '**Run Mode'** drop down menu in the toolbar.

In the '**Variable'** column (main window), you should get a drop-down menu of the available design variables. Select VDD as the parametric sweep variable. Note, only the design variables in the 'Design Variables' section of the ADE will be available in the drop-down list along with the system variable *temp*. So, make sure you have copied the design variables from the schematic into the ADE using the **Menu: Variables → Copy From Cellview…** command.

The '**Value**' column will list the nominal value of the parametric variable as set up in the ADE, in this case VDD will be 2.5V. In the '**Sweep**' column, make sure the option box is checked to enable the sweep.

In the '**Value List**' column enter the list of values – separated by spaces - for the VDD values that you want to simulate. For VDD +/-10%, this would be 2.25V, 2.5V, 2.75V. See Figure 11.

(4.3.4) The parametric analysis is run from the parametric analysis window not from the ADE window. In the Parametric Analysis window, run the '**Run Selected Sweeps**' command by clicking on the green play button. The simulation should run properly and both the text output window as well as the waveform output window should open automatically.

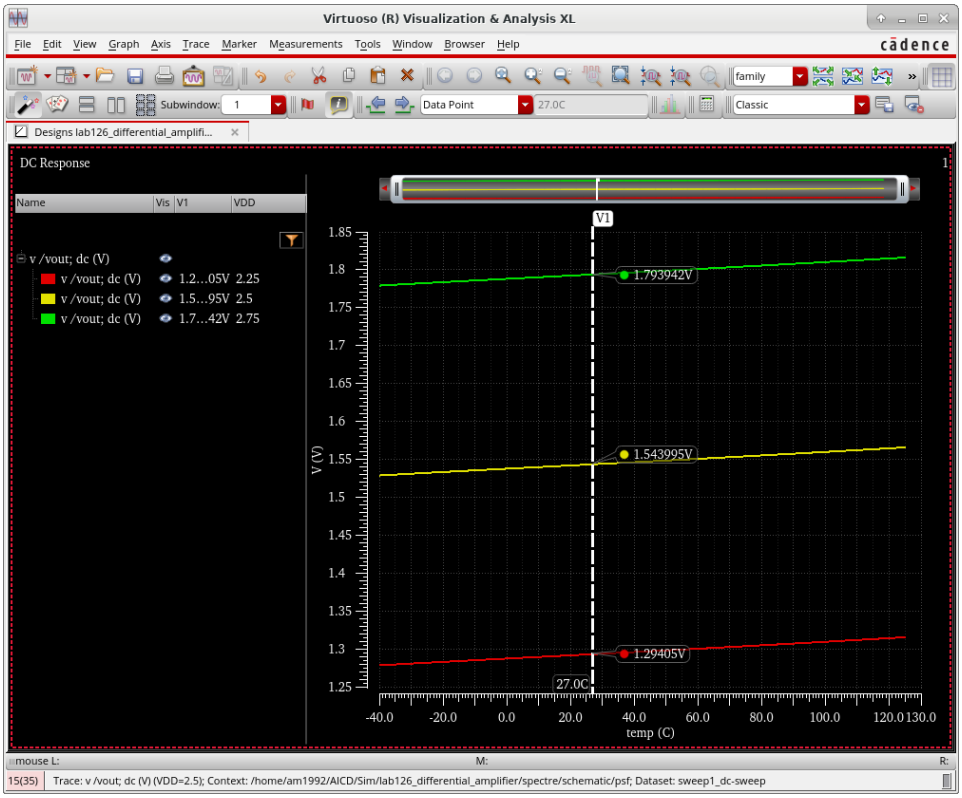
With the above settings, the ADE will perform a swept-DC analysis of *temp* from -40°C to 125°C, a total of 3 times – one simulation for each value of *VDD* (2.25V, 2.5V, 2.75*V*). See Figures 12 and 13.

(4.3.5) **Discussion**. When we add the effect of VDD variation (+/- 10%) to the temperature variation (-40 to 125°C), the DC operating point at v_{out} falls to 1.28V at its lowest (at VDD=2.25V, Temp=-40°C, and Process=NN) and rises to 1.81V at its highest (at VDD=2.75V, Temp=125°C, and Process=NN). This gives a total variation of around 0.5V over the full range of temperature and VDD variation. Under nominal conditions v_{out} is 1.55V (at VDD=2.5V).

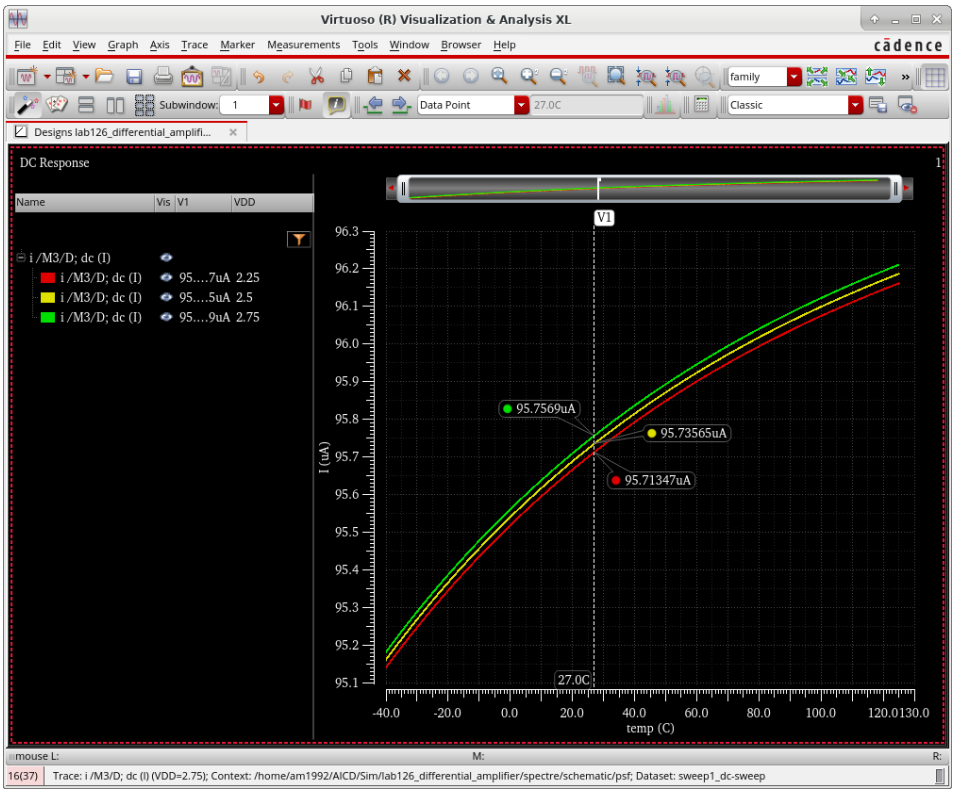
We want the DC output voltage to remain at approximately mid-rail for maximum signal-swing (VDD/2). As a ratio of vout/VDD we get 1.55V/2.5V for the nominal case, and 1.28V/2.25V and 1.81V/2.75V for the corner cases. The DC output voltage remains close to 55% to 65% of VDD in all cases (relatively close to mid-rail, 50% of VDD).

The current mirror loaded TCA shows a satisfactory level of robustness to changes in temperature and supply voltage over a wide range. However, whether we can say that this level of robustness is acceptable will depend on the required specifications for the application.

Note, we have not yet introduced any variation of the process parameters.



< Figure 212. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V >



< Figure 213. DC operating point I_{TAIL} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V >

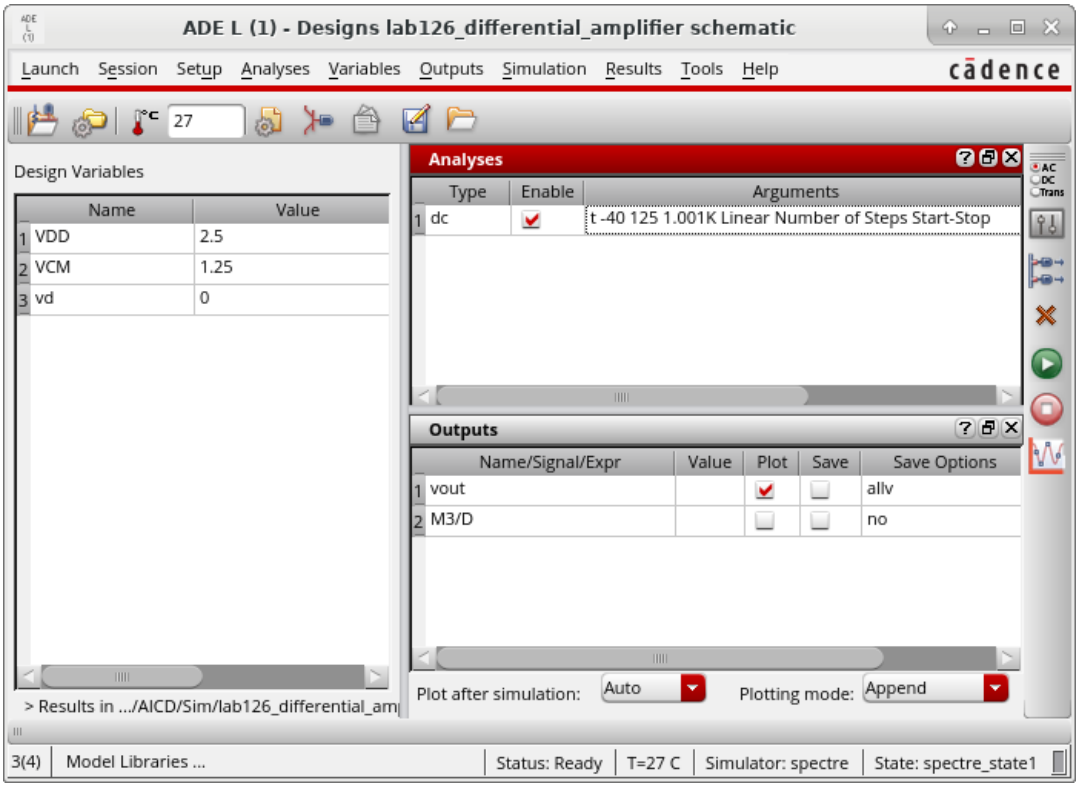
4.4 Simulating the Amplifier at Process Corner, Supply Voltage, and Temperature (PVT)

(4.4.1) We will setup a parametric analysis so we can observe the ***simultaneous*** effect of process (NN, SS, FF, SF, FS), voltage (2.25, 2.5V, 2.75V), and temperature (-40°C to +125°C) – PVT variation – on our circuit.

(4.4.2) While there are ways of automating the changing of process corners, we will be doing this manually (conducting multiple runs of the simulator) once for each process corner (SS, SF, FS, FF) plus nominal NN. This will require five parametric sweeps in total. Since this can be quite repetitive, we will add the outputs we wish to observe – the output net vout and the tail current node I_{TAIL} – to the ADE.

(4.4.3) From the ADE, select **Menu: Outputs → To Be Plotted → Select On Schematic** to enter the plotting mode of the ADE**.** Select the output net vout and the drain node current of the tail device (M3/D) for plotting. For now, only select vout for plotting. Note that your transistor names may be different – it depends on the order in which you instantiated the devices.

Make sure the Plotting Mode is set to 'Append'. This will ensure the output plots are added to the waveform window automatically on each run of the simulator. **Press the 'ESC' key to exit the plotting mode of the ADE**. See Figure 14.

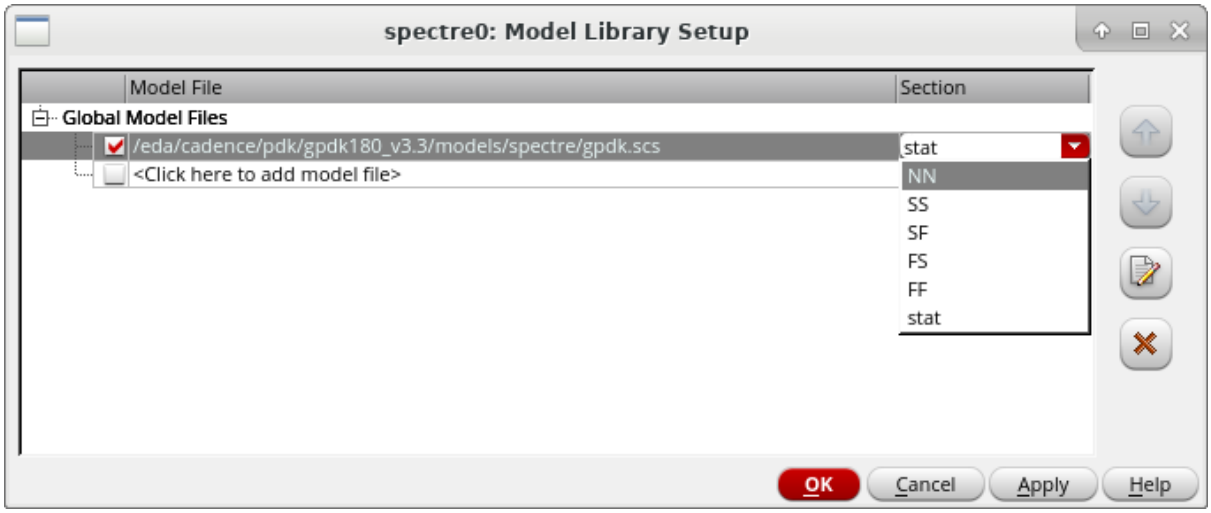


< Figure 214. The ADE with the Design Variables, Analyses and Outputs. >

Setting the Model Library Sections

(4.4.5) The Model Library file contains the model files that describe the NMOS and PMOS devices during simulation. As mentioned in the introduction, the device library models are divided into sections. There are typically six sections – stat, NN, SS, FS, SF, and FF that group together the model parameter variations according to their effect on NMOS and PMOS transistor speed.

(4.4.6) Model library files settings. From the ADE, go to **Menu: Setup → Model Libraries** to invoke the Model Library Setup form. See Figure 15. The main column lists the model files loaded by the Spectre simulator for use in our simulations (Recall that Spectre is the name of the SPICE simulator that the ADE runs). In our case, this is the GPDK.scs model file for the GPDK180 technology library (the '.scs' extension stands for 'spectre circuit simulator'). Note that this is the main model file; it links to several other files in the technology library (e.g., cmos.scs, diode.scs, bipolar.scs, capacitor.scs, resistor.scs, …).



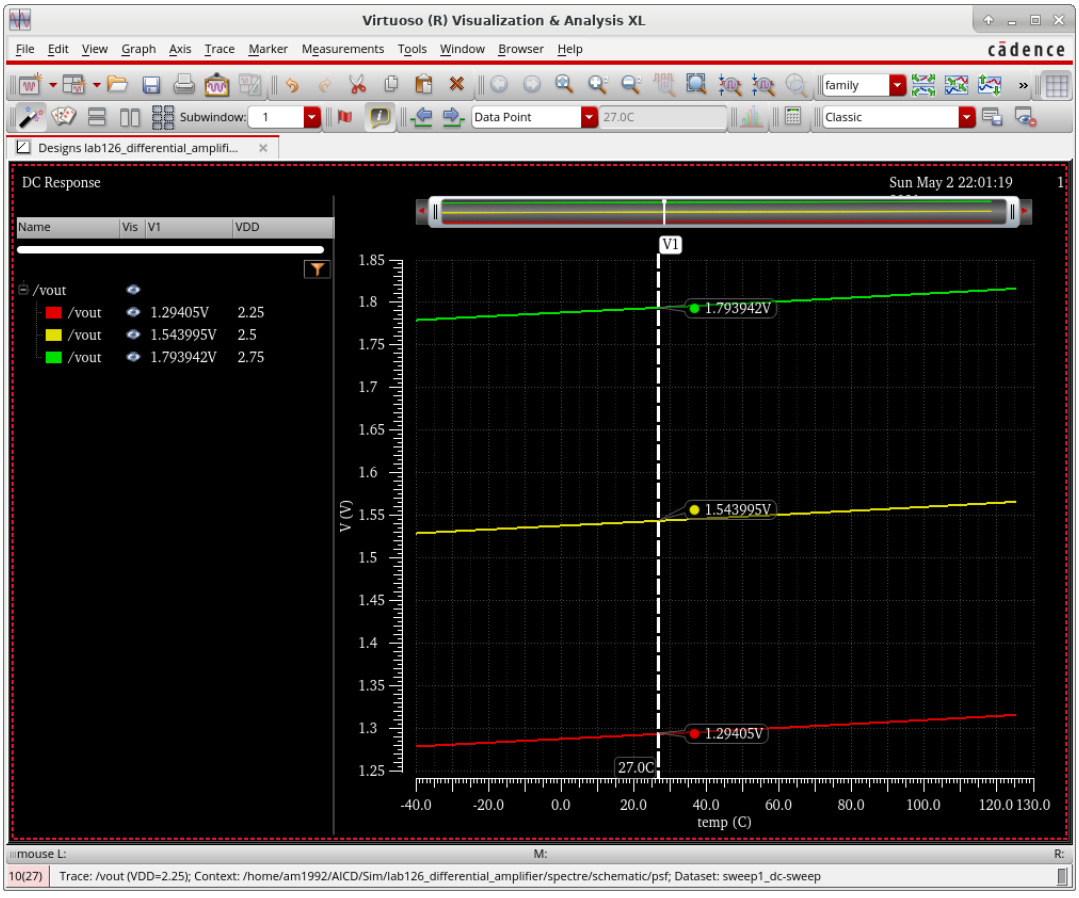
< Figure 215. The Model Library Setup Form. >

(4.4.7) In the 'section' column, alongside the model file, is a drop-down menu that lists the sections of the model file.

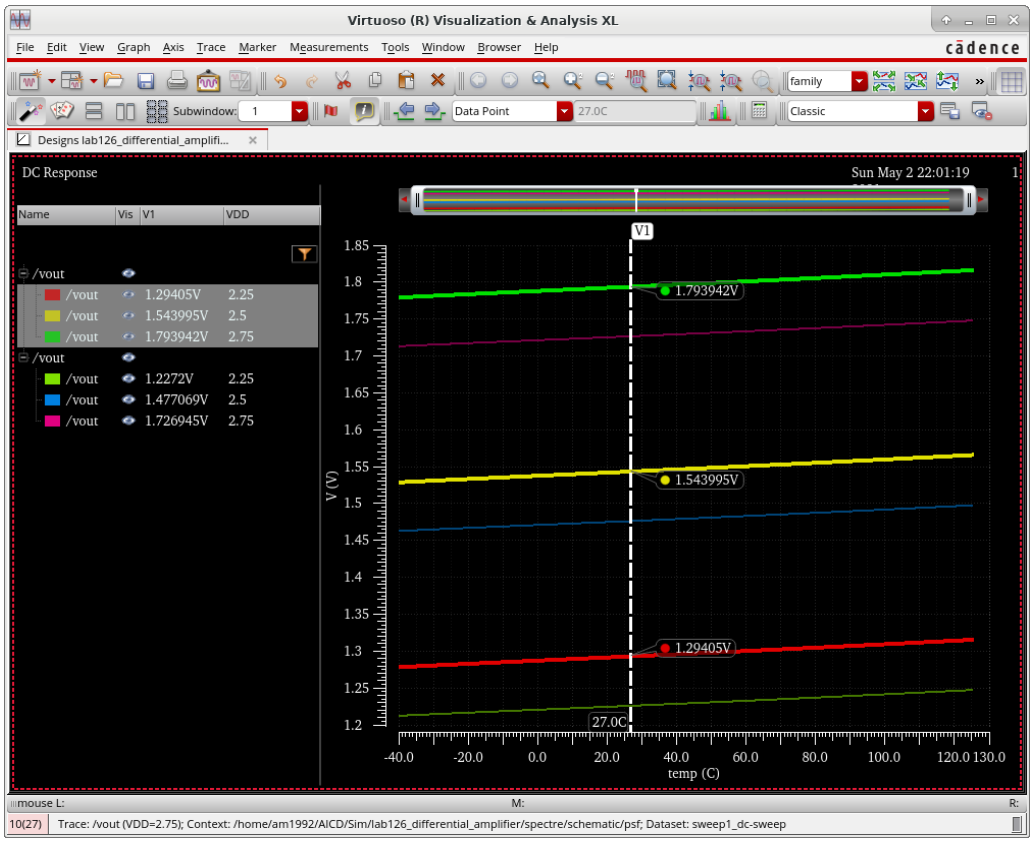
By default, this is set to **stat** – this is used for statistical analyses/ simulations (such as a Monte-Carlo analysis). However, to make use of the statistical analysis section requires additional functionality to instruct the simulator how the model parameters should be varied. This is available in the ADE-XL version of the ADE, but is not available in the basic ADE-L version which we use in this course. By default, **stat** behaves identical to the **NN** (nominal) option if no information on how to vary the parameters is supplied.

(4.4.8) Set the process section to **NN** (nominal/nominal) for the typical-mean performance. Apply or OK the form. From the Parametric Analysis window, run the '**Run Selected Sweeps**' command by clicking on the green play button. The waveform window should open and plot the output net vout for the **NN** process parameter case, across a temperature sweep -40°C to 125°C, at the voltage corners for VDD +/-10% (2.25V, 2.5V, and 2.75V).

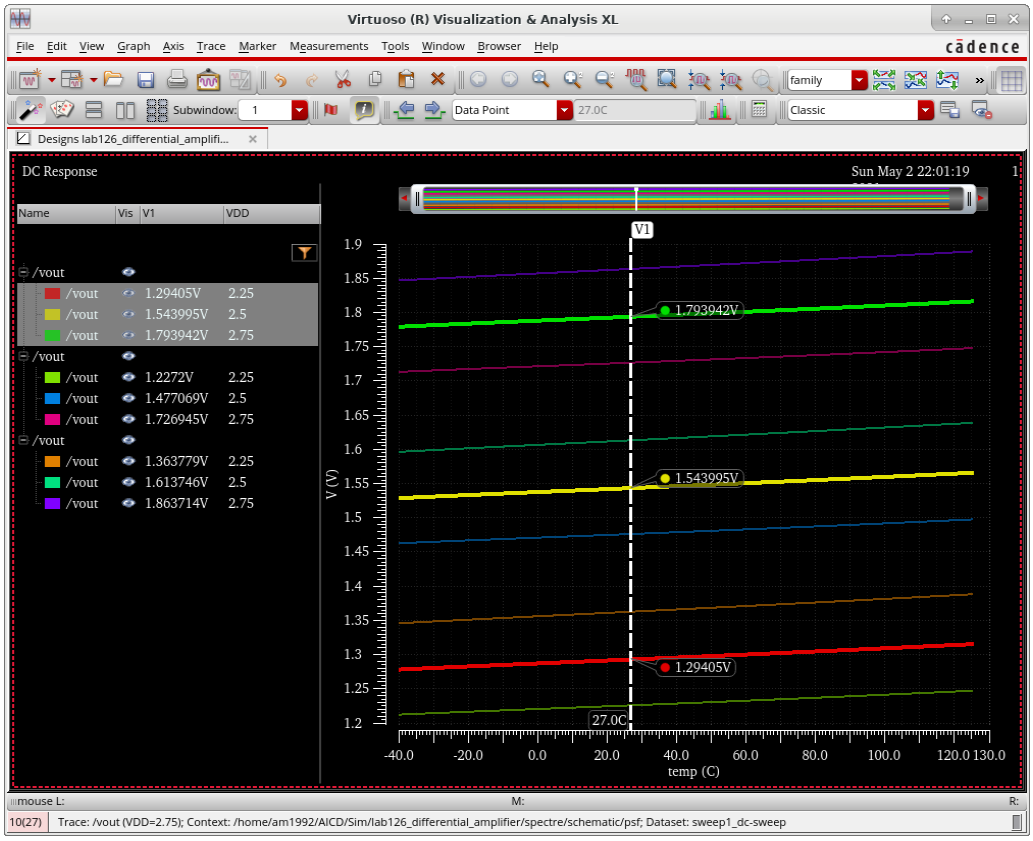
(4.4.9) Repeat the previous step for all process corners **SS**, **FF**, **SF**, **FS** (the order is not important here). For each corner, change the process section, apply or OK the form, and re-run the simulation. **Remember to run the simulation from the parametric tool not the ADE.** Ensure your ADE plotting mode is set to 'Append' so that each previous corner simulation is retained.



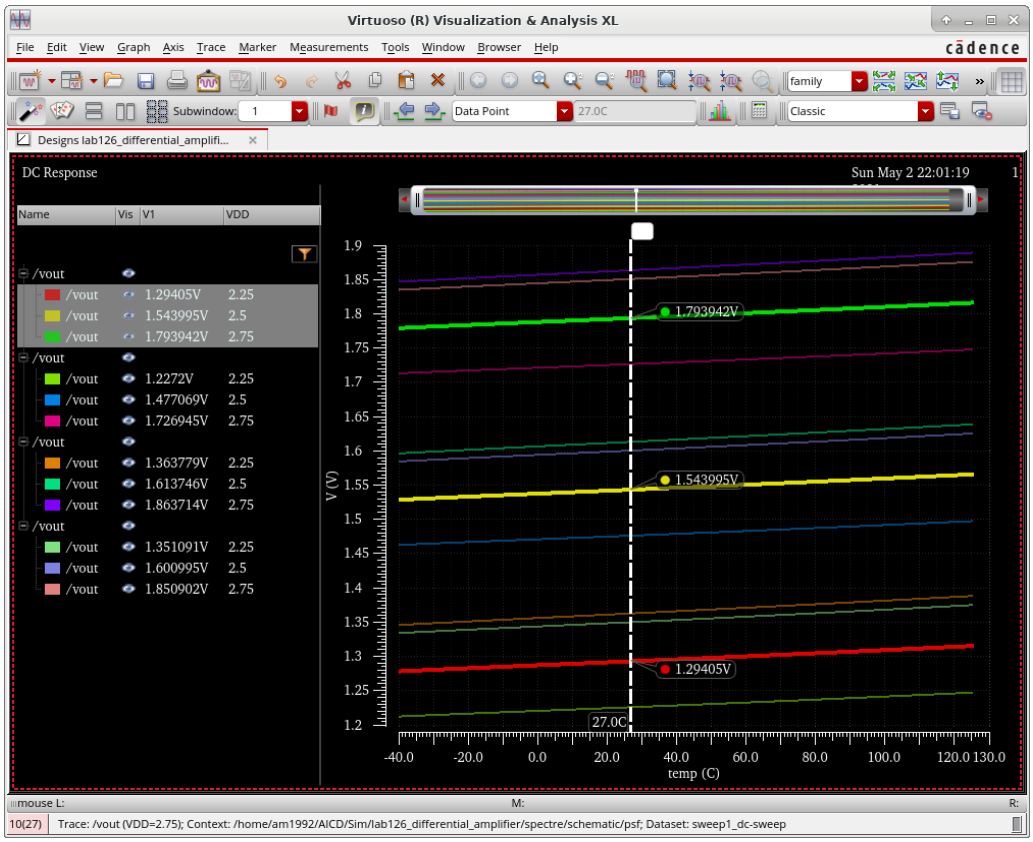
< Figure 216. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. For NN Process Section. >



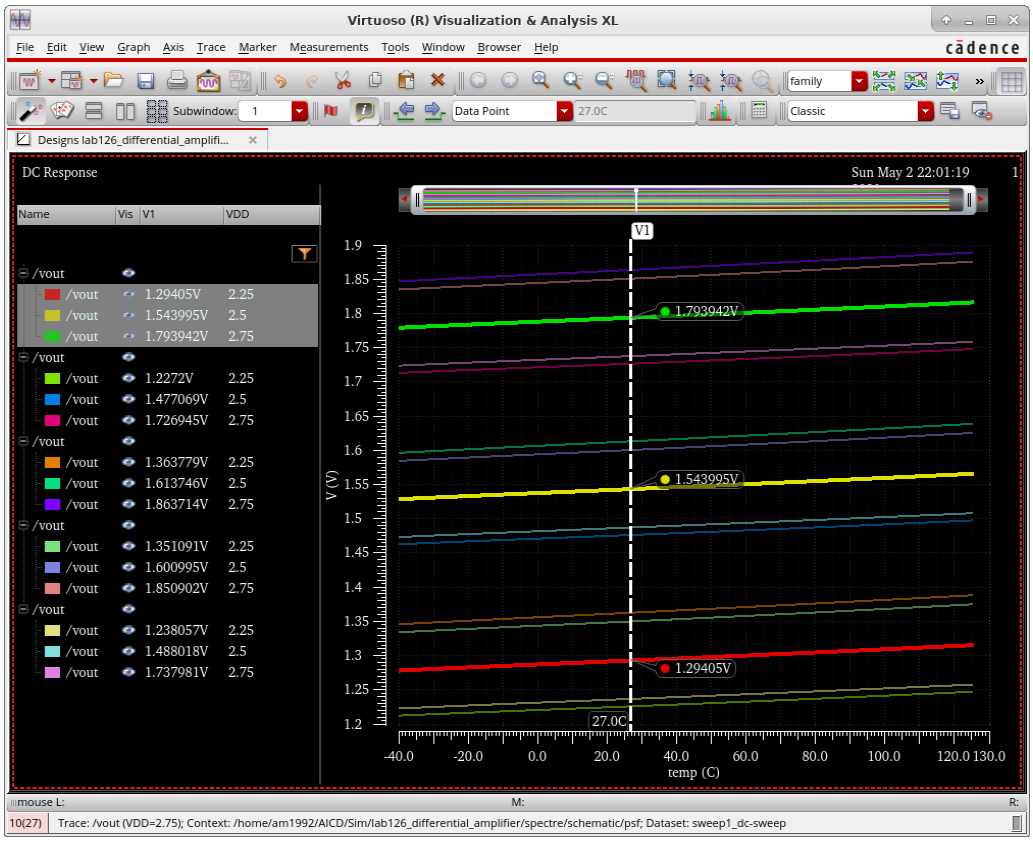
< Figure 217. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. For NN+SS Process Sections. >



< Figure 218. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. For NN+SS+FF Process Sections. >

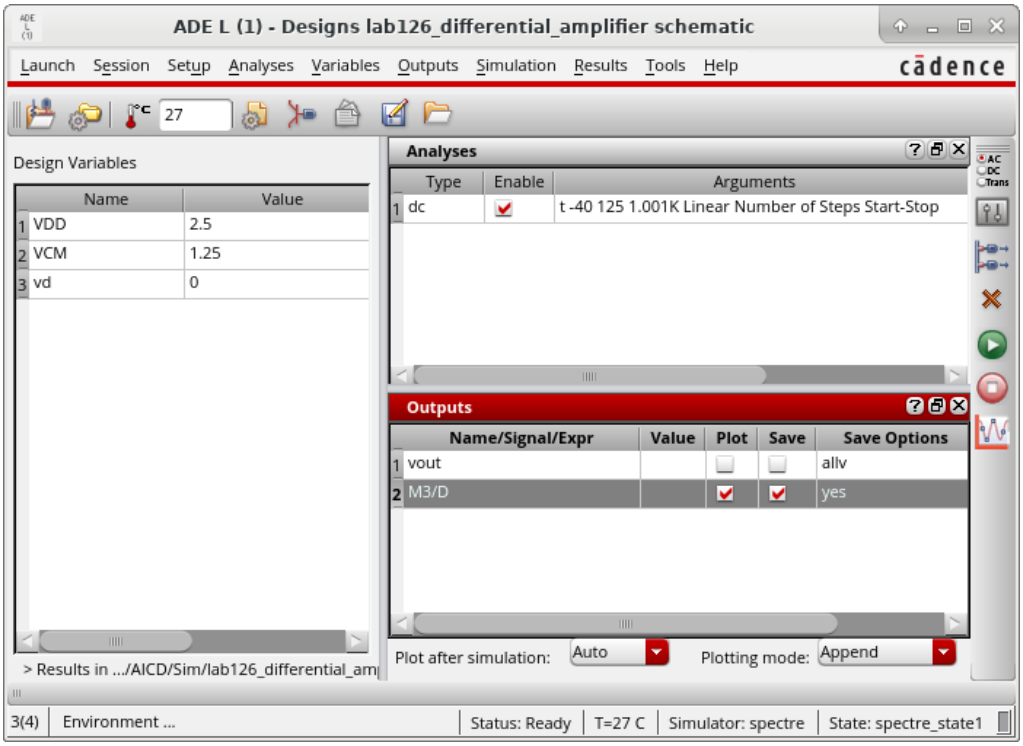


< Figure 219. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. For NN+SS+FF+SF Process Sections. >

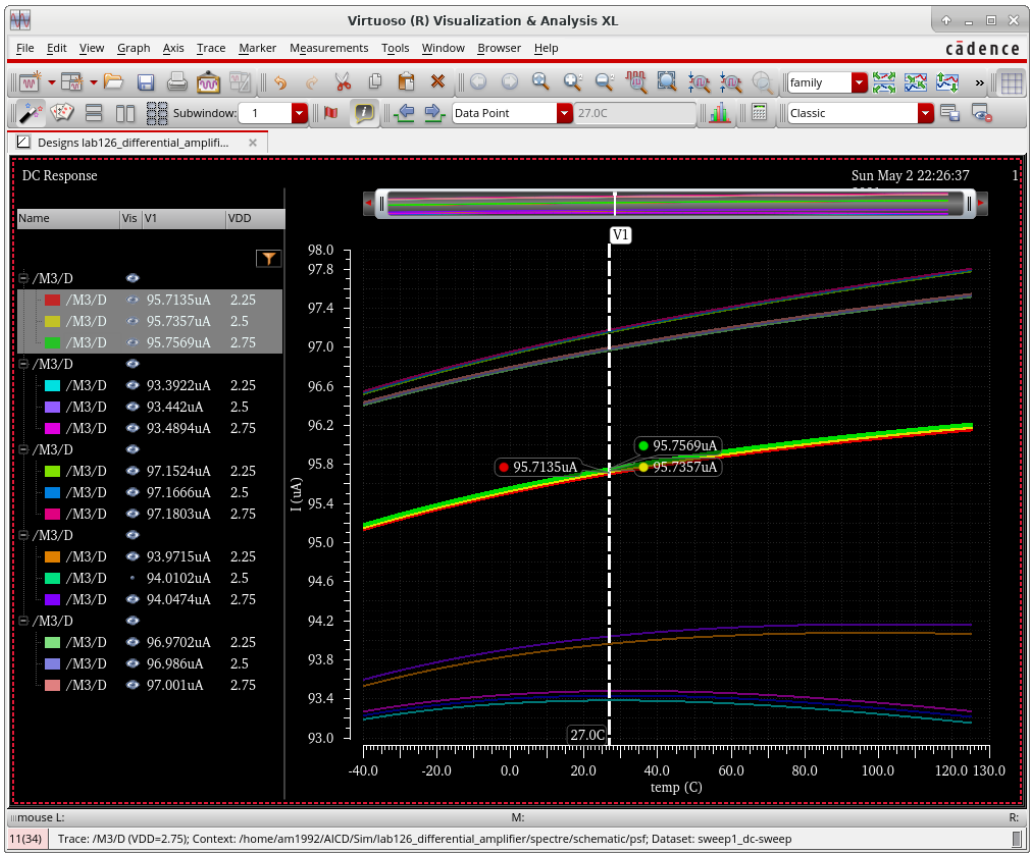


< Figure 220. DC operating point v_{out} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. For NN+SS+FF+SF+FS Process Sections. >

(4.4.10) Repeat the process again for the tail current (M3/D) output.



< Figure 221. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 222. DC operating point I_{TAIL} of the TCA for temperature -40°C to +125°C for VDD = 2.25V, 2.5V, and 2.75V. NN+SS+FF+SF+FS Process Sections. >

(4.4.11) **Discussion**. When determining the worst-case PVT variations (the corner cases), we are looking for the highest and lowest values of the circuit parameter under investigation across all the generated curves/plots. These can (literally) be found in the corners of the plot window.

For the circuit to have 'passed' the corner analysis, that is, deemed sufficiently robust to PVT variations, the highest and lowest values (i.e., the worst-case values) of the circuit parameter (v_{out}) must be within the range given in the specifications. If a parameter falls outside the specifications, the circuit has failed the test/analysis.

In the above simulations, over the full PVT variation (Figure 21), the DC operating point at v_{out} varies from 1.54V at nominal (VDD=2.5V, Temp=27°C, and Process=NN), to 1.21V at its lowest corner-case (VDD=2.25V, Temp=-40°C, and Process=SS), to 1.88V at its highest corner-case (VDD=2.75V, Temp=125°C, and Process=FF).

For the tail current (Figure 23), we get 95.7uA at nominal (VDD=2.5V, Temp-27°C, and Process=NN), 93.1uA at its lowest corner-case (VDD=2.25V, Temp=125°C, and Process=SS), and 97.8uA at its highest corner-case (VDD=2.75V, Temp=125°C, and Process=FF).

This gives a total v_{out} variation of ~ 1.54V +/- 0.34V and a tail current variation of ~ 95.7uA +/- 3uA. This would be considered acceptable in most applications (though, this would depend on the exact specifications of application). We see that the current mirror loaded TCA circuit is inherently self-biasing and robust to PVT (helped by the large NMOS/PMOS device sizes 10um/1um). Other circuits such as the differential amplifier with PMOS current source loads are very susceptible to variations in PVT and require an active feedback approach to maintain the proper bias.

In the next part of this lab, we will examine the role of feedback as a way of mitigating the adverse effects of PVT variation.

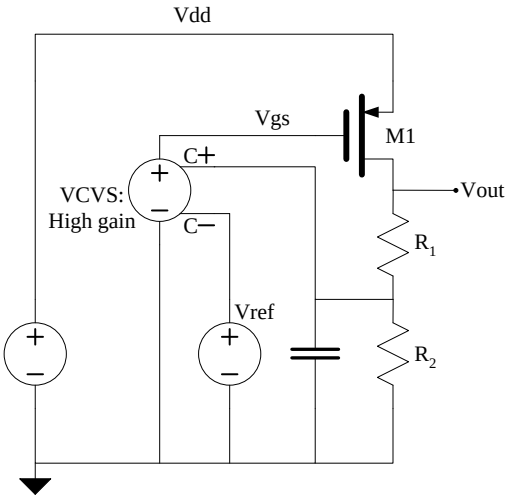
5. Circuit Simulation Exercises: Part 2

With guidance, you will:

- Examine the use of feedback to remove the problems associated with PVT variation.

5.1 The Controllable Voltage Reference Circuit

(5.1.1) Create the schematic for the IC voltage reference circuit in Figure 24. A summary of the circuit's operation is given in section 4.6 of the background material.



< Figure 223. IC Controllable Voltage Reference Circuit

Reproduced from Figure 30, section 4.6 of the background material. >

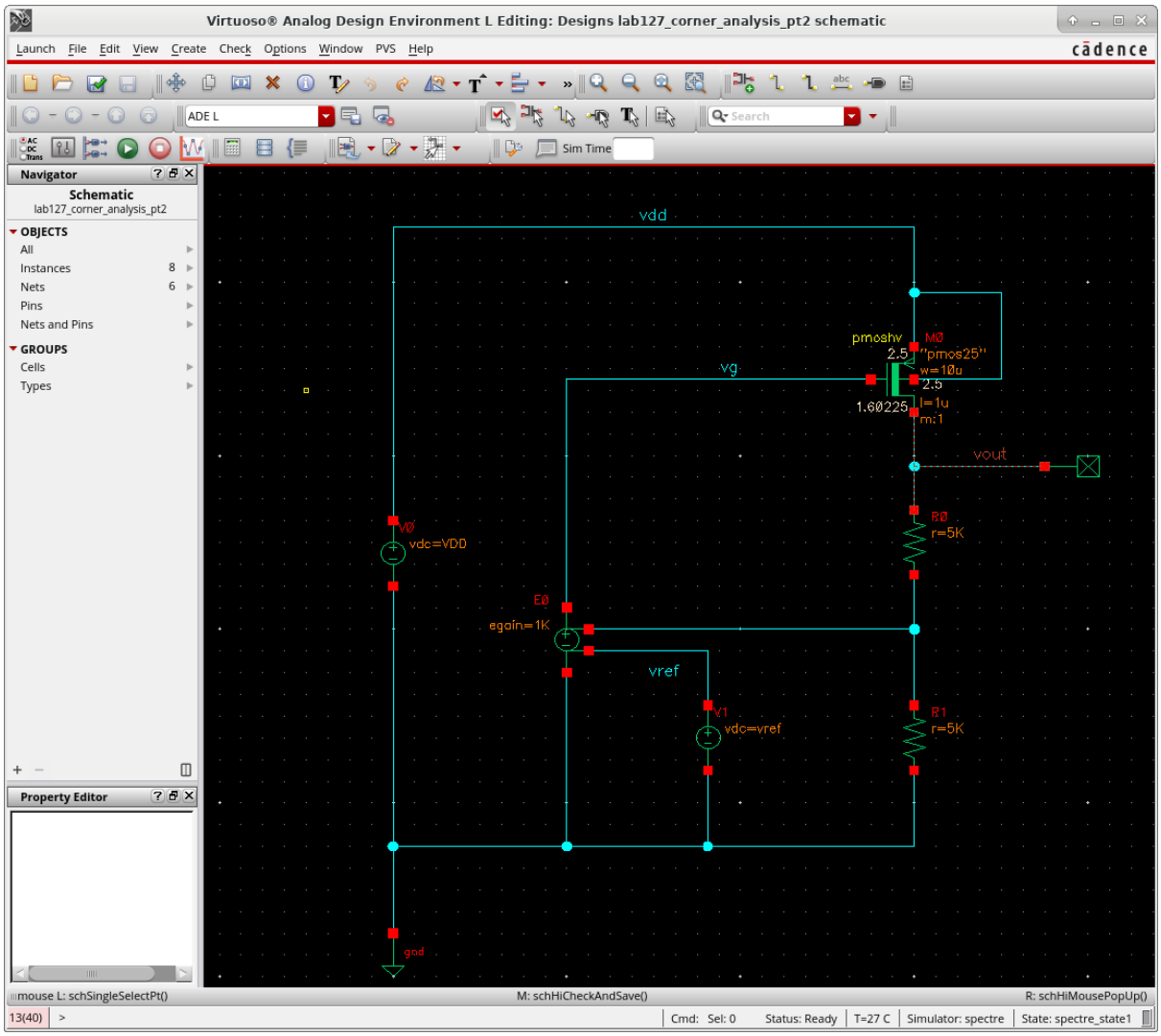
(5.1.2) You can either create the schematic from scratch (New File) or reuse the schematic of this circuit we created in lab 12.4, part 2 (lab124_vregulator_pt2). If creating a new file or copying the old cell, give the schematic a meaningful name e.g., lab127_corner_analysis_pt2.

(5.1.3) In the schematic editor window, construct the circuit given in Figure 24 (25). From the analogLib library instantiate and place two DC voltage sources (**vdc**), two resistors (**res**), a voltage-controlled voltage source (**vcvs**), and a ground reference (**gnd**). Also instantiate one high-voltage PMOS FET (**pmoshv**) from the GPDK180 library.

Parameterise your design. For the DC supply voltage, set the voltage attribute to the string **DC Voltage** = VDD. For the input, set the reference to the string **DC Voltage** = vref.

Dimension the PMOS device 10um/1um (**width** = 10u, **length** = 1u). Set the VCVS to a gain of 1K (**Voltage Gain** = 1K). Set the two resistors to 5KΩ (**Resistance** = 5K). Since we are simulating with DC you can ignore the capacitor. Leave all other attributes unchanged.

(5.1.4) Wire the components as in Figure 24 (25) using the **Wire (narrow)** command. **Remember to connect the body/bulk terminal of the PMOS device to the most positive net in the circuit (VDD=2.5V)**. Using the **Create Wire Name** command, name the DC supply net vdd, the PMOS gate net vg, and the reference input net vref.



< Figure 224. Controllable Voltage Reference Circuit.

(5.1.5) Check and save your schematic using the **Check and Save** command. Remember to use the output area of the CIW to diagnose any problems with your schematic and/or simulations if there are any warning or error messages when checking, saving, or simulating your work.

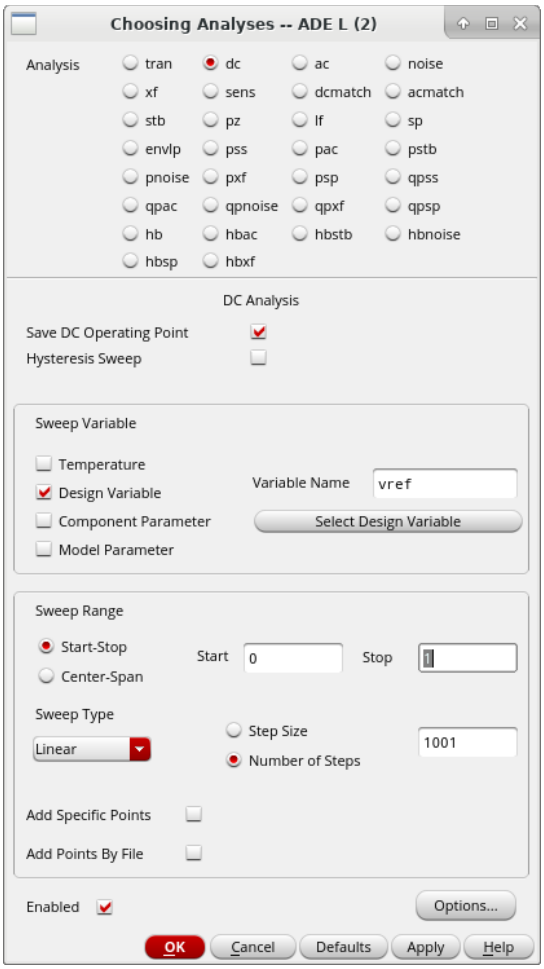
5.2 DC Sweep Analysis of the Volage Reference Circuit

(5.2.1) We will simulate the voltage reference using a DC sweep. We will sweep the input reference vref from 0 to 1V and observe how accurately the output vout tracks the input. From the Schematic Editor window, launch the Analogue Design Environment (ADE) from **Menu: Launch → ADE L**.

(5.2.2) Copy the design variables from the schematic into the ADE. From the ADE window, choose **Menu: Variables à Copy From Cellview**. You should now see the VDD and vref variables in the 'Design Variables' list of the ADE. Initialise VDD = *2.5*V and vref = *0.8*V.

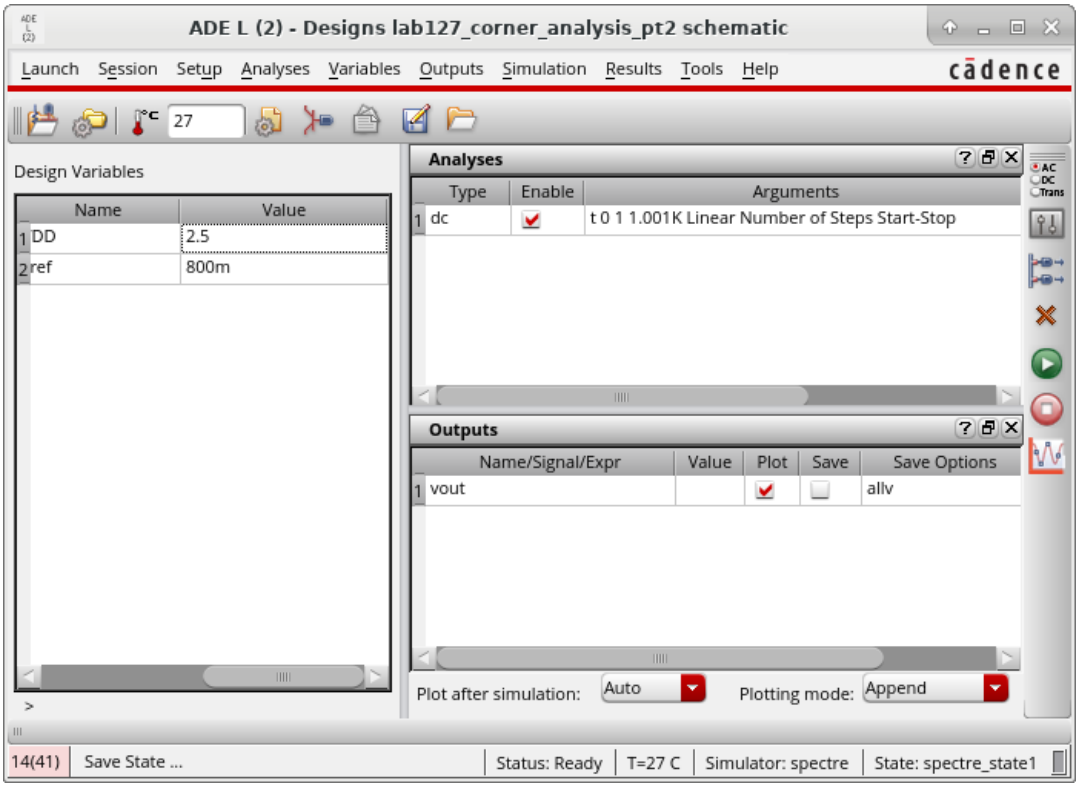
(5.2.3) From the ADE window, go to **Menu: Analyses à Choose…** or use the **Choose Analyses** button on the right-side of the ADE window (top button) to invoke the 'Choosing Analyses – ADE L' form. Select the '**dc**' option button.

In the choosing analyses form. Check the 'Save DC Operating Point' box. Check the 'Design Variable' box as the sweep variable. Type vref as the variable name. Choose the 'Start-Stop' option button with Start = 0 and Stop = 1 for the sweep range. Select 'Linear' from the drop-down menu as the sweep type. Choose the 'Number of Steps' option button with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 26. OK the form. The **dc** analysis should now appear in the 'Analyses' list of the ADE window.

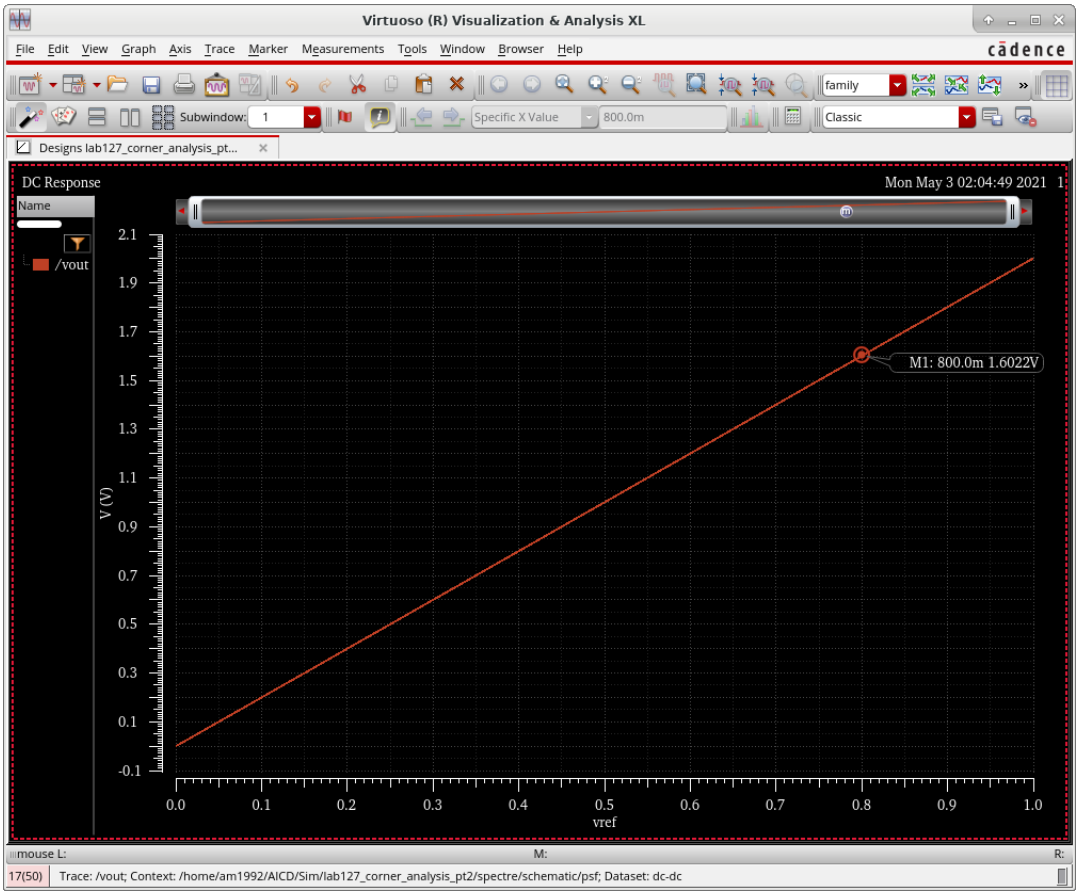


< Figure 225. The Choosing Analyses Form. >

(5.2.4) From the ADE, select **Menu: Outputs → To Be Plotted → Select On Schematic** to enter the plotting mode of the ADE**.** Select only the output net vout for plotting. **Make sure the Plotting Mode is set to 'Append'**. This will ensure the output plots are added to the waveform window automatically on each run of the simulator. **Press the 'ESC' key to exit the plotting mode of the ADE**.



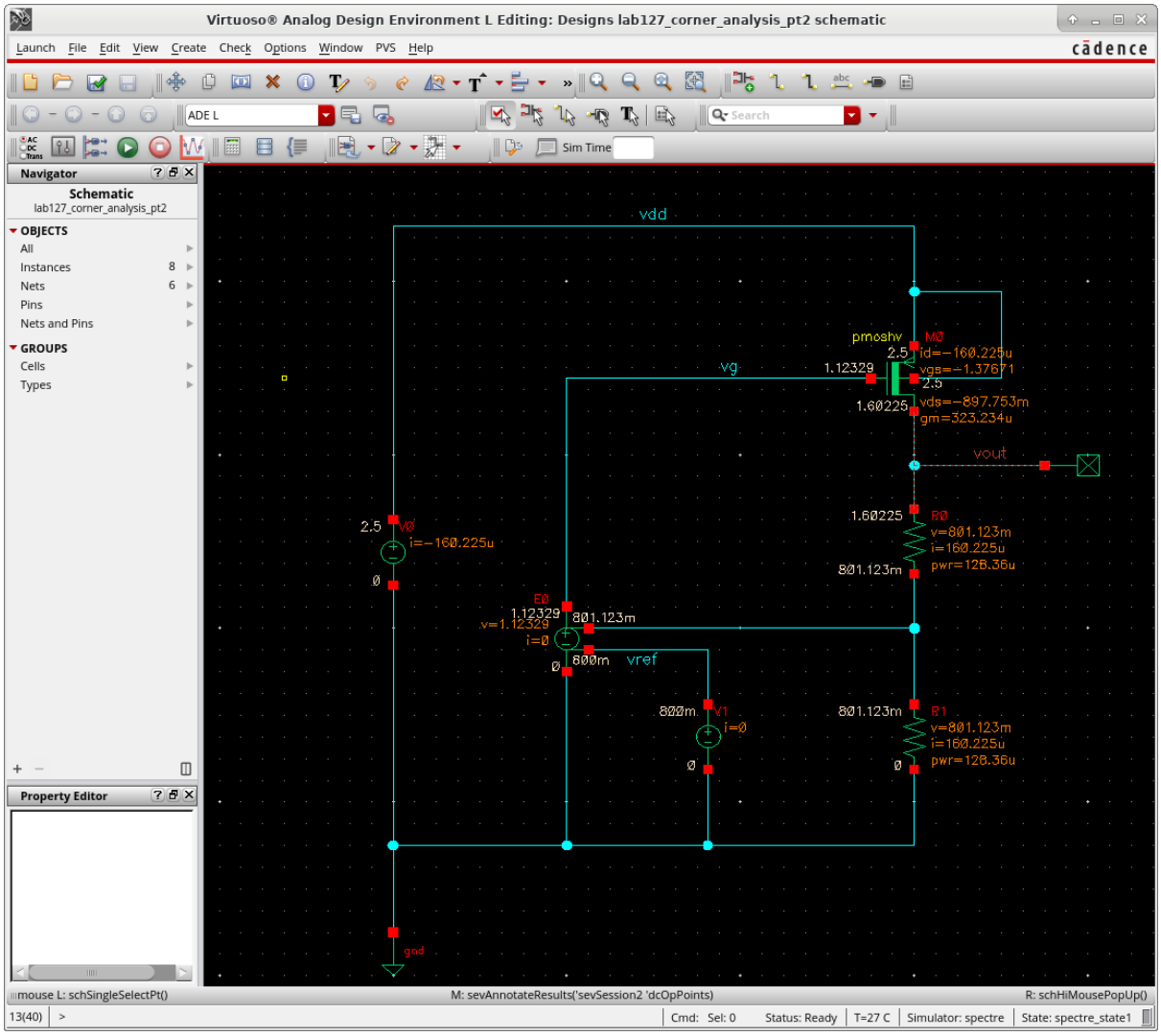
< Figure 226. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 227. Output Voltage vout Tracking as input reference is swept from 0 to 1V. >

(5.2.5) In the Analog Design Environment window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error. The text output window should appear but there will be no waveform window.

(5.2.6) From the ADE window choose **Menu: Results à Annotate à DC node voltages…** If you now look in the schematic window, you should see the node voltages annotated directly on the schematic. Again, from the ADE window, choose **Menu: Results à Annotate à DC operating points…** You should now see (along with the node voltages) additional DC operating point information annotated on the schematic including the DC bias currents. See Figure 29.



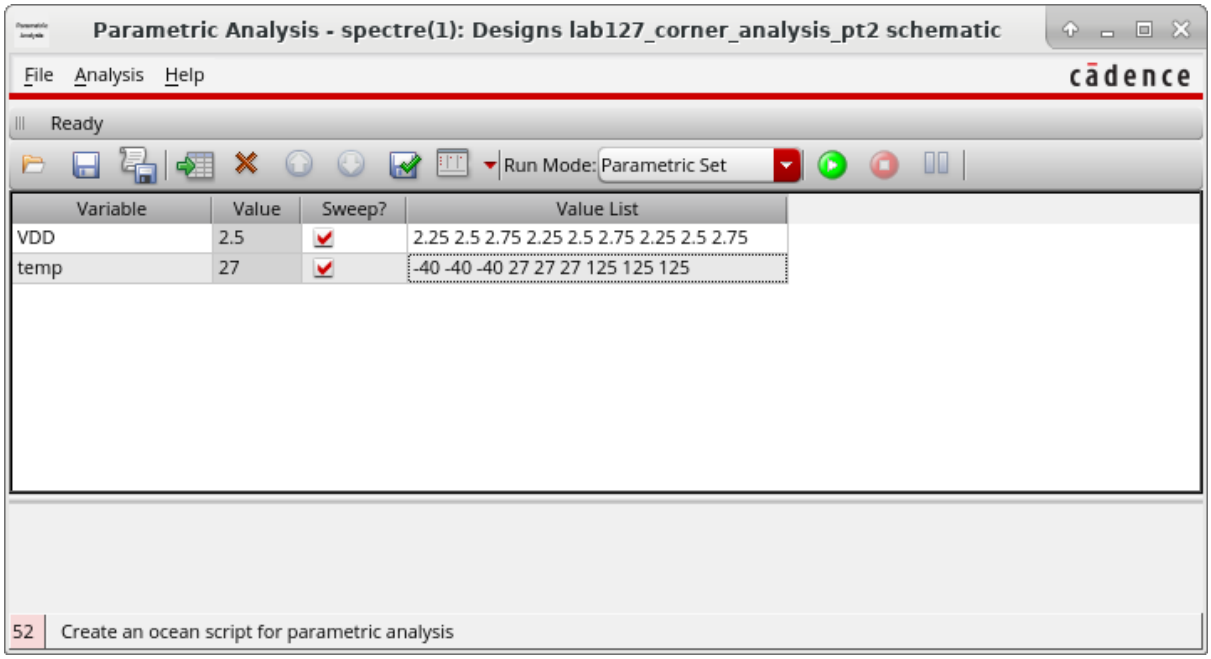
< Figure 228. Controllable Voltage Reference Circuit with Node Voltage and DC Operating Point Annotations. >

(5.2.7) **Discussion**. Refer to section 4.6 of the background material for a discussion on circuit operation. The important point to note is that the high gain of the VCVS regulates the output voltage(vout) to be twice that of the voltage reference (vref). The factor of two occurs due to the even division of the feedback signal across the potential divider. The feedback acts to maintain the same voltage on both inputs to the VCVS, this requires an output voltage across the resistors equal to 2*vref so that vcvs(+) equals vcvs(-). The high gain of the VCVS is used to overcome any non-ideal behaviour of the PMOS – or any other disturbance – such as PVT variation. The higher the gain the more closely the output will track the voltage reference (by a factor of two.) Note that the PMOS device adds an inversion in the loop – so the input to the VCVS is the negative terminal and the feedback signal goes to the positive terminal.

5.3 DC Sweep Analysis of the Volage Reference Circuit over Process, Voltage and Temperature (PVT)

(5.3.1) We will setup a PVT corner analysis so we can observe the ***simultaneous*** effect of process (NN, SS, FF, SF, FS), voltage (VDD +/-10%), and temperature (-40°C to 125°C) as we sweep the input reference from 0 to 1V.

(5.3.2) Since we are sweeping the input reference voltage vref from within the ADE itself – we will use the parametric analysis tool to vary the power supply voltage (VDD – 2.25V, 2.5V, and 2.75V) and temperature (-40°C, 27°C, and 125°C). Note that here we are using the nominal and corner values of both voltage and temperature (we are not sweeping the temperature linearly over the full range as we were before). See Figure 30. See (5.3.3) below for an explanation of how the value list is set up.



< Figure 229. Parametric Analysis Tool. Set up for a corner combination of VDD (+/- 10%) = 2.25V, 2.5V, and 2.75V and Temperature -40°C, 27 °C, and 125°C) >

(5.3.3) Note that when setting up the parametric list/set in the parametric analysis tool, you have to explicitly set the combinations of VDD and temperature. **The parametric tool will not itself iterate over all combinations.** The number of items/values in each list must match and each pair of values in the lists will be simulated in the order listed. For example, in Figure 30, the combination of {VDD, temp} simulated by the parametric tool will be:

{2.25, -40°C },

{2.5V, -40°C },

{2.75V, -40°C },

{2.25V, 27°C },

{2.5V, 27°C },

{2.75V, 27°C },

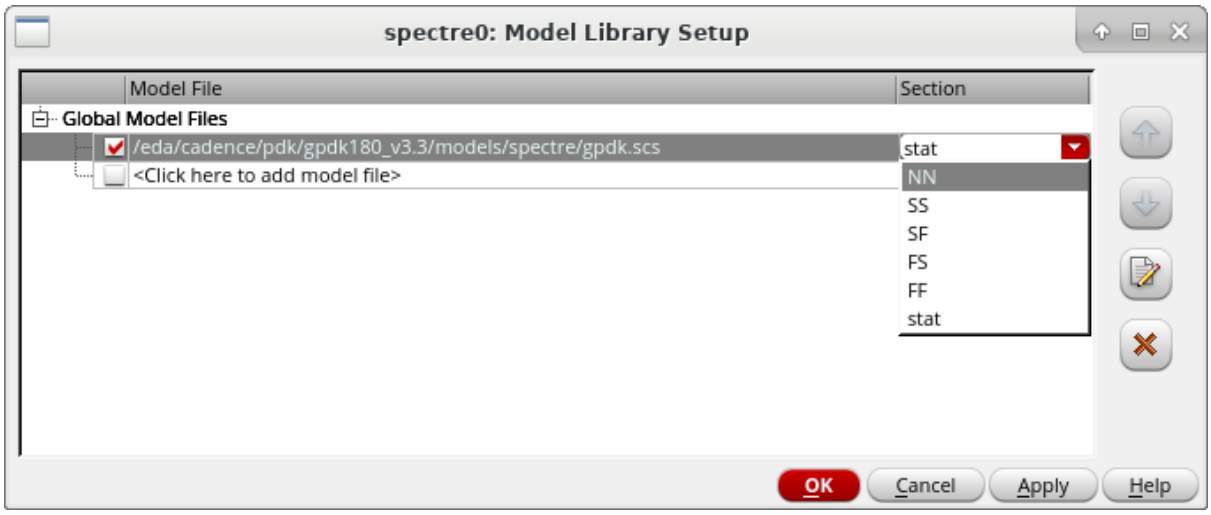
{2.25V, 125°C },

{2.5V, 125°C },

{2.75V, 125°C }.

This covers all corner combinations of VDD and temperature.

(5.3.4) Model library files settings. From the ADE, go to **Menu: Setup → Model Libraries** to invoke the Model Library Setup up form. Set the process section to **NN** (nominal/nominal) for the typical-mean performance. From the Parametric Analysis window, run the '**Run Selected Sweeps**' command by clicking on the green play button. The waveform window should open and plot the output net vout for the **NN** process parameter case, across temperature corners (-40°C, 27°C, and 125°C) at the VDD voltage corners (2.25V, 2.5V, and 2.75V).

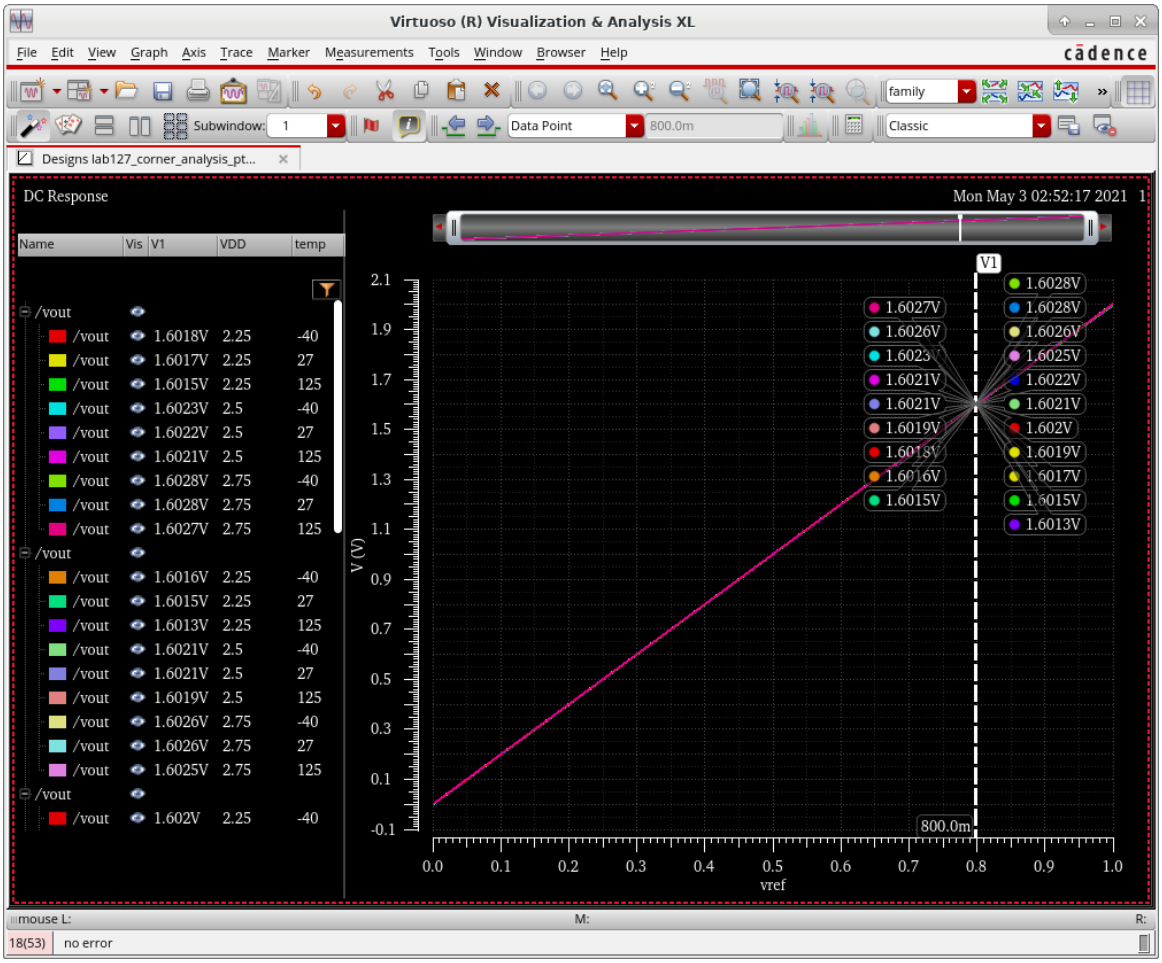


< Figure 230. Section Selection Drop-down Menu in the Model Library Setup Form. >

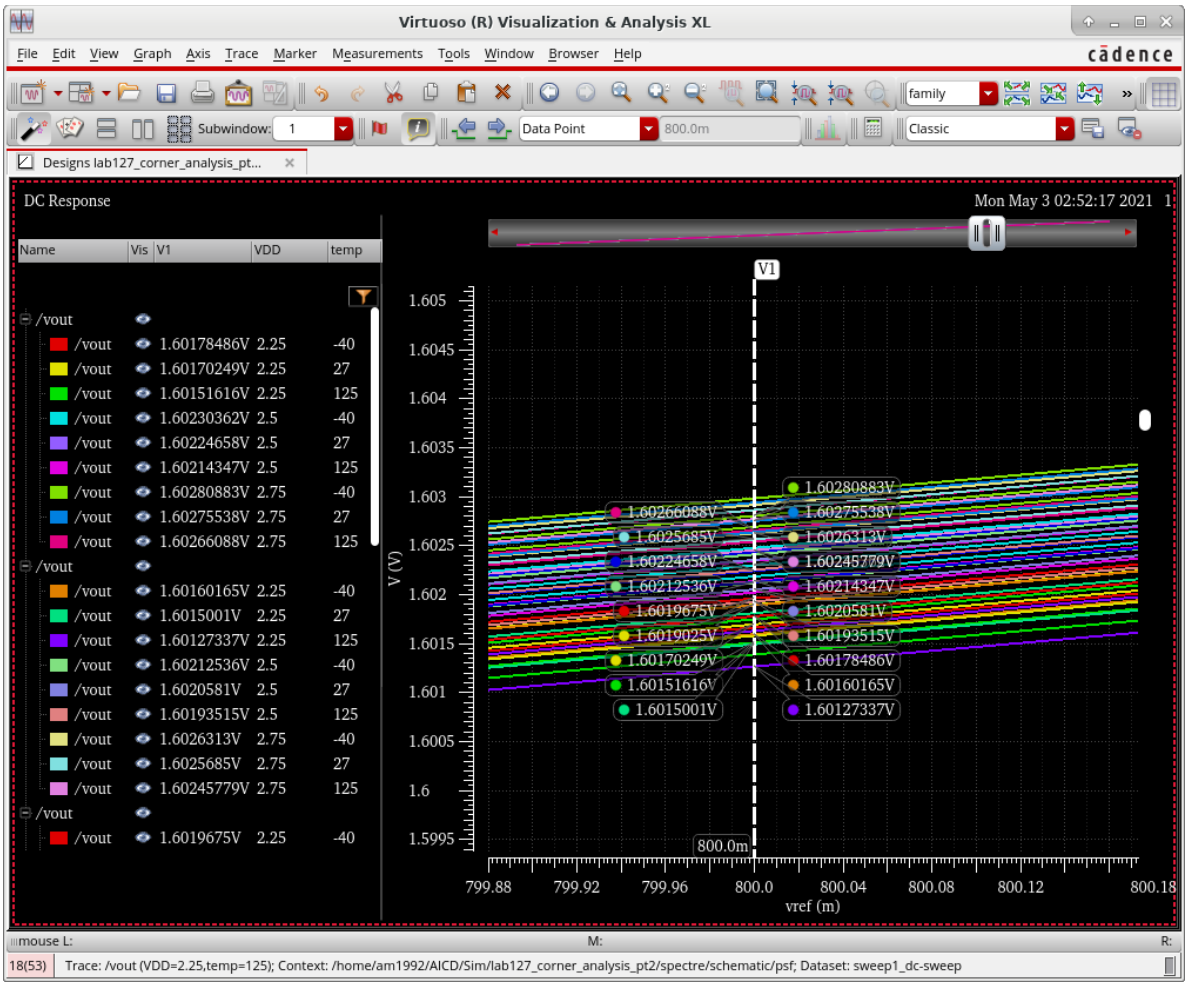
(5.3.5) Make sure the plotting mode is set to **Append**. Repeat the previous step for all process corners **SS**, **FF**, **SF**, **FS** (the order is not important here). For each corner, change the process section, apply or OK the form, and re-run the simulation. **Remember to run the simulation from the parametric tool not the ADE.**

(5.3.6) **Discussion**. Referring to Figures 32 and 33. With the feedback amplifier set to a gain of 1K we have an accurate tracking of the input voltage reference (multiplied by 2), even in the face of full PVT variation. For an input refence of 0.8V, the worst-case the output voltage is 1.6028V. This represents a tracking accuracy of:

$$T.A = \left(1 - \frac{Vout - (2 * Vref)}{(2 * Vref)}\right) \times 100\% = \left(1 - \frac{1.6028V - 1.6V}{1.6V}\right) \times 100\% = 98.25\%$$



< Figure 231. DC operating point v_{out} of the voltage regulator. Temperature -40°C, 27 °C, and +125°C. VDD = 2.25V, 2.5V, and 2.75V. NN+SS+FF+SF+FS Process Sections. VCVS Gain=1K>



< Figure 232. As above. Close-up around 0.8V. Temperature -40°C, 27 °C, and +125°C. VDD = 2.25V, 2.5V, and 2.75V. NN+SS+FF+SF+FS Process Sections. VCVS Gain=1K>

(5.3.7) Repeat steps 5.3.1 to 5.3.5 for a feedback amplifier **VCVS** gain of 100K.

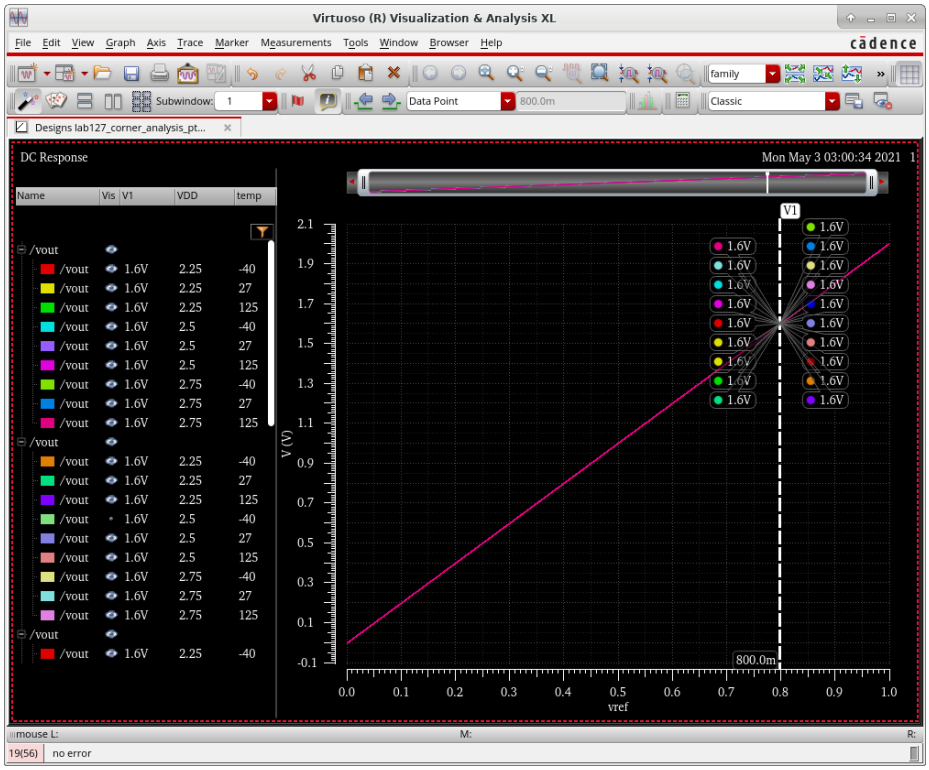
(5.3.8) **Discussion**. Referring to Figures 33 and 34. Increasing the feedback gain to 100K, the worst-case output voltage is 1.600028V. We therefore increase the tracking accuracy to,

$$T.A = \left(1 - \frac{1.600028V - 1.6V}{1.6V}\right) \times 100\% = 99.9825\%$$

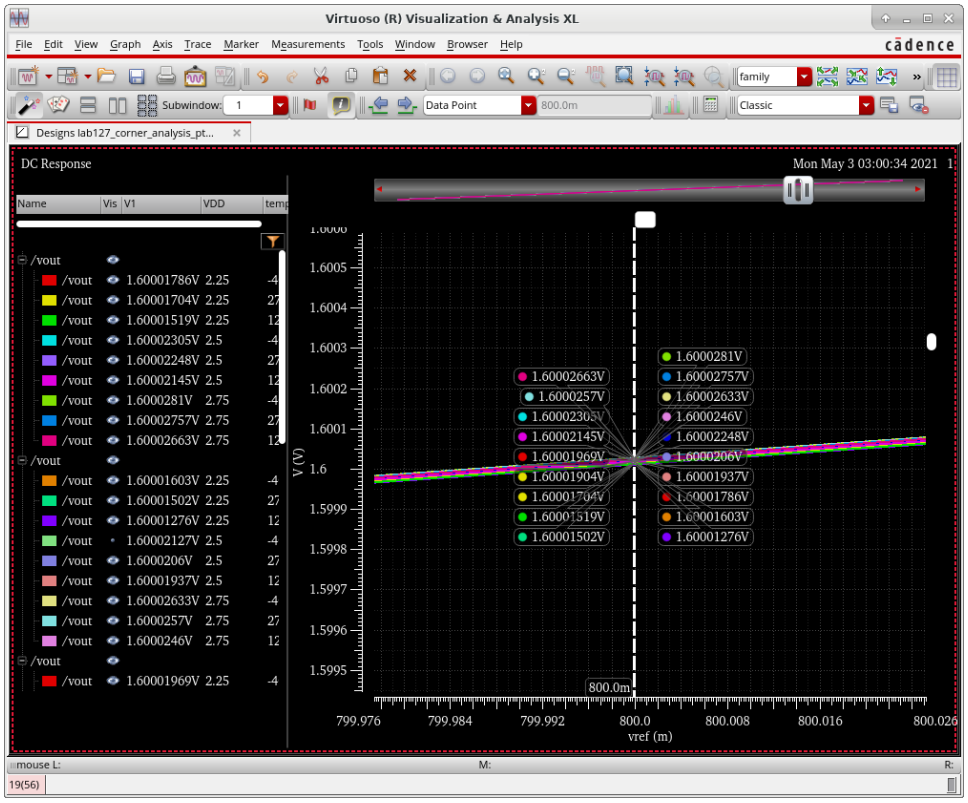
It is not surprising that increasing the gain by a factor of x100 improves the tracking accuracy by the same amount. The higher the gain the better the tracking accuracy and the less impact PVT variation has on circuit performance.

Note from lab 12.4, however, that there is a trade-off between tracking accuracy and stability – the higher the gain the worse the stability margin.

Analogue design is about trading one performance metric for another.



< Figure 233. DC operating point v_{out} of the voltage regulator. Temperature -40°C, 27 °C, and +125°C. VDD = 2.25V, 2.5V, and 2.75V. NN+SS+FF+SF+FS Process Sections. VCVS Gain=100K>



< Figure 234. As above. Close-up around 0.8V. Temperature -40°C, 27 °C, and +125°C. VDD = 2.25V, 2.5V, and 2.75V. NN+SS+FF+SF+FS Process Sections. VCVS Gain=100K> **End of Lab!**